

Recording-Aware WiFi CSI-Based Human Activity Recognition: A Baseline Towards
WiFi-DensePose

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by

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Abstract

This report presents the second-stage experimental development of a WiFi sensing project motivated by the WiFi-DensePose framework. The internship began with a preliminary technical study of WiFi-based human sensing, Channel State Information (CSI), DensePose, the public WiFi-DensePose implementation and its Docker-based simulation mode. The present work advances that study by analysing a real recorded CSI dataset and constructing a reproducible human activity-recognition baseline. The dataset contains fifteen recordings across three classes—empty, standing and moving. Each valid CSI packet contains 130 parsed numeric values, of which the final 124 values are treated as 62 I/Q pairs. These are converted into subcarrier amplitudes, temporally windowed, normalized using training-only statistics and represented through statistical features. Several conventional machine-learning models are compared under ordinary and recording-aware validation. A feature-based Random Forest reaches 91.23% accuracy on held-out test windows, while recording-aware validation is substantially lower. An RBF-SVM and PCA-SVM are selected through recording-aware cross-validation. On the final untouched recordings, the selected PCA-SVM obtains 57.89% window-level accuracy and 66.67% recording-level accuracy using mean decision-score aggregation. The results demonstrate that coarse activity information is present in CSI while also exposing strong recording/session variation. The work establishes a practical baseline and identifies generalisation, richer CSI processing and synchronized ground truth as key requirements for future progression toward WiFi-based dense human pose estimation.

Keywords—WiFi sensing, Channel State Information, CSI, human activity recognition, device-free sensing, Random Forest, Support Vector Machine, PCA, WiFi-DensePose.

I. Introduction

A. Background and Motivation

Human sensing has traditionally relied on cameras, depth sensors, wearables and other dedicated sensing devices. Although these approaches can provide rich observations, they may depend on line of sight, favourable illumination, physical contact or dedicated hardware. The preliminary technical study undertaken by the authors identified these limitations as motivation for investigating radio-frequency sensing, particularly WiFi CSI, as a non-visual sensing modality.

The preliminary study specifically examined the WiFi-DensePose direction. It reviewed human pose estimation, device-free sensing, CSI, OFDM, multipath propagation, phase sanitization and the architecture of a publicly available WiFi-DensePose implementation. It was explicitly positioned as an architectural and simulation investigation rather than a real-world accuracy evaluation; real CSI acquisition, dataset collection, model retraining and field deployment were identified as later phases.

B. From WiFi-DensePose Study to CSI Activity Recognition

The present internship phase follows that roadmap. Rather than directly attempting dense pose estimation without first establishing a local signal-processing pipeline, the project uses recorded CSI to build a controlled activity-recognition problem. The three classes—empty, standing and moving—provide a first test of whether the locally available CSI recordings contain repeatable information about human presence and movement.

The preliminary report proposed a future dataset containing synchronized CSI, RGB video, DensePose annotations and environmental metadata, and identified environmental variability, multipath, limited datasets, synchronization and domain adaptation as major research challenges. The present experiment addresses a smaller part of this problem by examining real CSI recordings and testing generalisation across complete recordings.

C. Foundational WiFi-DensePose Work

The foundational motivation for the project was the open-source WiFi-DensePose implementation supplied by the project mentor, Dr. Partha Pakray. The preliminary report records that the framework can operate in simulation mode, process CSI, expose REST/WebSocket interfaces and produce higher-level outputs such as occupancy and pose-related visualisations. [1].

The academic direction is consistent with earlier work such as Person-in-WiFi [3], which demonstrated fine-grained person perception from WiFi, and DensePose From WiFi [2], which investigated dense body correspondence using WiFi

amplitude and phase.

D. Problem Statement

Can a locally recorded WiFi CSI dataset be processed into a reliable machine-learning representation that distinguishes empty, standing and moving states, while evaluating generalisation using complete unseen recordings rather than randomly split overlapping windows?

E. Significance of the Present Study

The project does not claim a new classification algorithm. Its contribution is the establishment of an end-to-end, recording-aware CSI baseline connected directly to the WiFi-DensePose research trajectory. The experiments also reveal a substantial difference between random-window validation and recording-level validation, providing evidence that session-specific characteristics can inflate apparent performance.

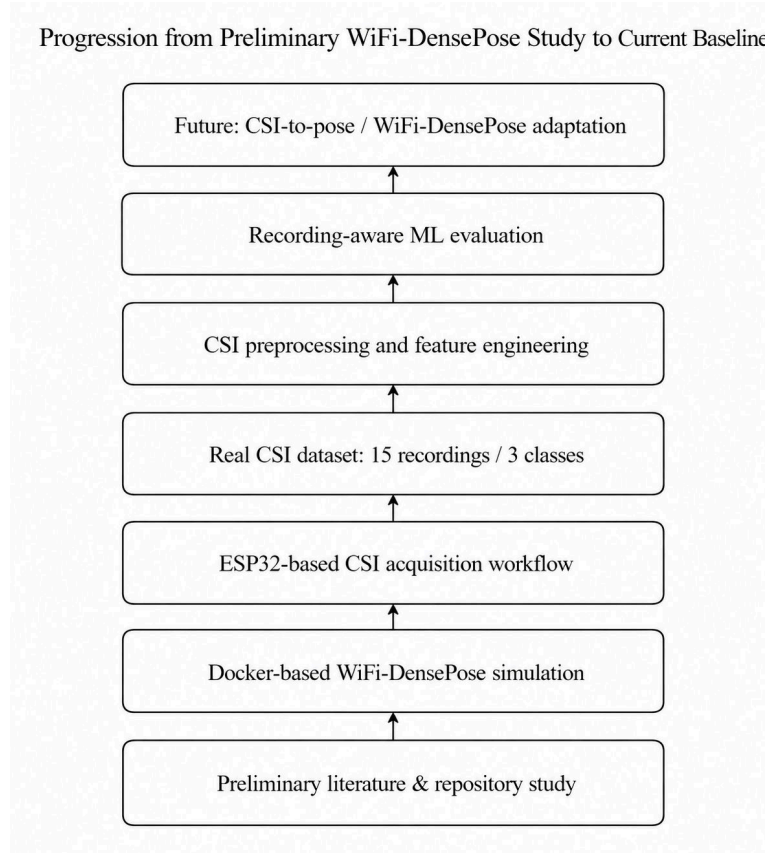


Figure 1.1. Progression from the preliminary WiFi-DensePose study to the present CSI activity-recognition baseline.

II. Objectives of Internship Program

The primary objective of the internship was to approach WiFi-based human sensing as a computer science and intelligent-systems problem rather than as a purely signal-processing exercise. The work therefore began by establishing the computational pipeline that connects raw wireless measurements to machine-readable representations and finally to an inference task. In this context, the team studied Channel State Information, its relationship with OFDM subcarriers and multipath propagation, and the way human movement modifies the measured wireless channel.

A second objective was to understand the WiFi-DensePose framework supplied as the foundational research direction by the project mentor, Dr. Partha Pakray. The team examined how a WiFi sensing system can be organized into acquisition, preprocessing, feature extraction, deep-learning inference, pose estimation, tracking and visualization stages. The preliminary Docker-based study established familiarity with the repository architecture and demonstrated that the software could operate in simulation mode before specialized CSI hardware was available.

From a data-engineering perspective, the internship aimed to construct a clean and reproducible CSI processing pipeline. This involved inspecting the fifteen CSV recordings, parsing the serialized CSI field, validating packet structure,

identifying malformed measurements and converting valid I/Q observations into a consistent numerical representation. The project further required temporal windowing and normalization procedures that preserve the distinction between training recordings and unseen recordings. These steps were treated as part of the model-design problem because leakage or inconsistent preprocessing can produce misleading machine-learning results.

From a machine-learning perspective, the objective was to investigate which representation of CSI is most useful for coarse human activity recognition. The team therefore compared direct window-level CSI representations with engineered statistical descriptors covering both subcarrier distributions and temporal changes. Several conventional classifiers were evaluated, including Random Forest, Extra Trees, K-nearest neighbours, logistic regression, histogram-based gradient boosting and linear and RBF-kernel SVMs. The intention was not simply to maximize an accuracy number, but to understand how representation, classifier choice and dimensionality affect generalisation on a small wireless-sensing dataset.

A particularly important objective was to design an evaluation strategy appropriate for correlated time-series data. Because the project uses overlapping temporal windows, a naive random split can place highly similar portions of one physical recording into both training and validation partitions. The team therefore investigated recording-aware GroupKFold validation and reserved recording 05 of each class as a final unseen test set. This converts the evaluation from a conventional pattern-classification exercise into a more meaningful experiment on cross-recording generalisation.

Finally, the internship aimed to establish a technically defensible baseline for the next stage of the broader WiFi-DensePose research programme. The present system stops at three coarse states—empty, standing and moving—but the engineering decisions, data pipeline and validation strategy are intended to support future extensions toward richer human postures, synchronized visual ground truth, phase-aware CSI processing and eventually WiFi-based pose estimation. In that sense, the objective of the internship was not only to build a classifier, but to create a reliable experimental foundation on which more sophisticated CSI-to-pose research can be developed.

The scope is deliberately narrower than full dense pose estimation. The present three-class classifier does not estimate skeletons, DensePose UV coordinates or multi-person body correspondences.

III. Methodology

A. Overall Experimental Pipeline

The workflow consists of dataset inspection, CSI parsing, packet validation, I/Q extraction, amplitude calculation, temporal windowing, training-only normalization, statistical feature construction, classifier comparison, recording-aware model selection and final unseen-recording evaluation.

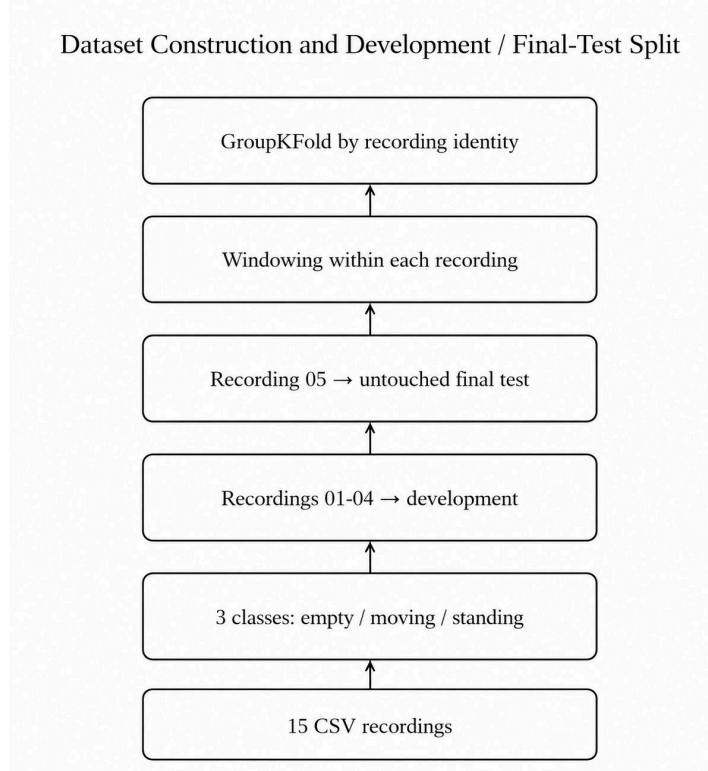


Figure 3.1. Dataset construction and separation of development and final-test recordings.

B. Preliminary WiFi-DensePose Simulation Phase

Before analysing recorded CSI, the authors studied and executed the WiFi-DensePose implementation in Docker on an Apple M3 MacBook Air using ARM64 macOS. In the absence of a physical CSI source, the software entered simulation mode.

The simulation was not used as an accuracy benchmark. It was used to understand software execution, CSI processing, inference flow and communication interfaces. The preliminary study observed simulated CSI packets, occupancy maps, pose-related outputs, vital-sign estimates, REST responses and WebSocket streams.

C. Experimental Hardware and Software Environment

The broader project used an ESP32-based CSI acquisition workflow for the locally recorded measurements, while the preliminary WiFi-DensePose implementation was examined in a Docker environment. The simulation stage was conducted on an Apple MacBook Air with an Apple M3 processor, ARM64 macOS, Docker, the WiFi-DensePose container image and network interfaces for HTTP, WebSocket and UDP communication. The CSI receiver/transmitter source files are retained in the supplied csi_receiver project archive.

Parameter	Configuration
Machine	Apple MacBook Air (M3)
Operating system	macOS, ARM64
Container platform	Docker
WiFi-DensePose image	ruvnet/wifi-densepose:latest
CSI acquisition platform	ESP32-based CSI workflow
Activity dataset	15 CSV recordings: empty, moving, standing
HTTP interface	Port 3000
WebSocket interface	Port 3001
CSI/UDP interface	Port 5005
ML environment	Python, NumPy, Pandas, Scikit-learn, Matplotlib

Table 3. Hardware and software configuration used across the preliminary and experimental phases.

D. CSI Dataset and Recording Organization

The dataset contains fifteen CSV recordings: five for each of three classes—empty, standing and moving. Each CSV contains timestamp, sample_number and csi_data. The filename encodes the class and recording number.

Recording	Class	Raw rows	Valid	Malformed
empty_01.csv	empty	626	625	1
empty_02.csv	empty	467	464	3
empty_03.csv	empty	443	442	1
empty_04.csv	empty	380	379	1
empty_05.csv	empty	1096	1096	0
moving_01.csv	moving	862	862	0
moving_02.csv	moving	1055	1053	2
moving_03.csv	moving	1147	1145	2
moving_04.csv	moving	1070	1070	0
moving_05.csv	moving	961	960	1
standing_01.csv	standing	764	764	0
standing_02.csv	standing	889	889	0
standing_03.csv	standing	983	983	0
standing_04.csv	standing	947	943	4
standing_05.csv	standing	1047	1046	1

Table 1. Recording-wise CSI data-quality summary reproduced from the experimental notebook.

Across the fifteen recordings, 12,737 raw rows were inspected. A total of 12,721 rows contained the expected 130 numeric values and 16 rows were rejected as malformed.

E. ESP32-Based CSI Acquisition Workflow

The recorded CSI dataset represents the practical transition from the software-simulation stage to physical wireless sensing. The project archive includes the ESP32 CSI receiver/transmitter workflow used to generate serialized CSI records that were stored as CSV files. Preserving packet order and recording identity allowed the subsequent Python analysis to treat each collection session as an independent experimental group rather than as an anonymous pool of windows.

F. CSI Packet Parsing

The notebook extracts signed integers from the serialized csi_data field. A valid packet is required to contain exactly 130 parsed numeric values. This criterion is applied uniformly to all recordings.

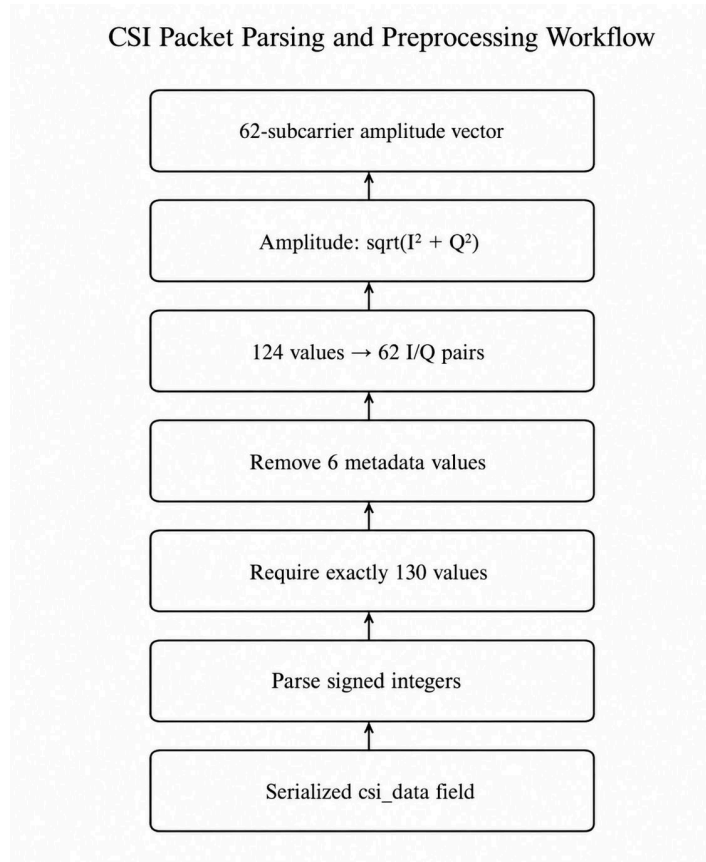


Figure 3.2. CSI packet parsing and validation workflow used in the activity-recognition baseline.

Figure: CSI Packet Representation Used in the Baseline

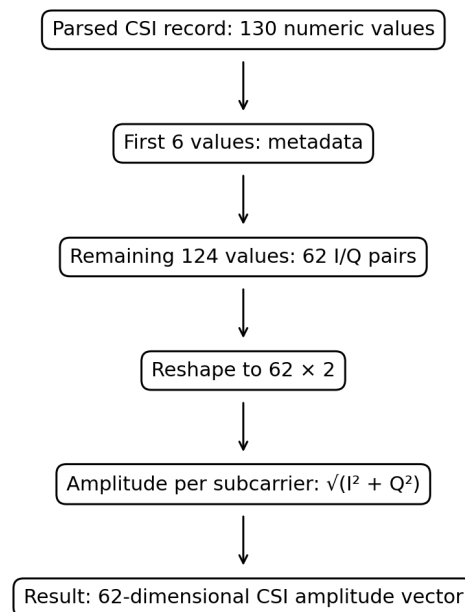


Figure 3. CSI packet representation used in the activity-recognition baseline.

G. I/Q Extraction and Amplitude

The first six parsed values are treated as metadata. The remaining 124 values are interpreted as 62 I/Q pairs, producing a 62×2 array.

$$H_k = I_k + jQ_k$$
$$A_k = \sqrt{I_k^2 + Q_k^2}$$

The first inspected clean sample produced a 62-element amplitude vector. Its first ten amplitudes were approximately 26.31, 27.07, 28.28, 28.44, 27.66, 26.25, 25.96, 27.29, 26.48 and 19.92.

H. Clean CSI Dataset

After filtering, the clean amplitude matrix has shape $12,721 \times 62$. The packet-level class distribution is 3,006 empty, 5,090 moving and 4,625 standing.

I. Training and Testing Protocol

Recordings 01–04 from every class are used for model development and recording 05 is held out for final evaluation. This produces 9,619 training packets and 3,102 test packets. Training contains 1,910 empty, 4,130 moving and 3,579 standing packets; testing contains 1,096 empty, 960 moving and 1,046 standing packets.

Recording-level separation is essential because adjacent CSI packets and overlapping windows from the same recording are correlated. A random split can therefore produce overly optimistic performance.

J. Temporal Windowing

The notebook forms windows of 100 consecutive CSI packets with a step of 50 packets. Each window has shape 100×62 . This produces 174 training windows and 57 test windows. Training contains 32 empty, 77 moving and 65 standing windows; testing contains 20 empty, 18 moving and 19 standing windows.

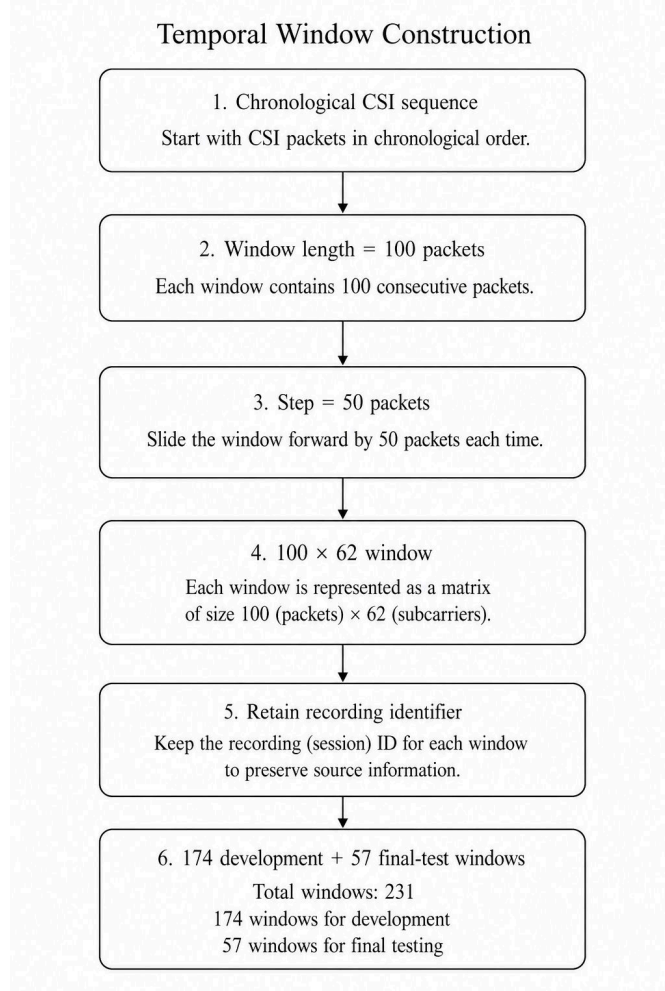


Figure 3.4. Construction of 100-packet temporal windows using a 50-packet step.

K. Training-Only Normalization

The mean and standard deviation for each subcarrier are calculated from the training tensor only and then applied to both train and test tensors.

$$X' = (X - \mu_{train}) / \sigma_{train}$$

The measured normalized training mean is -3.215378×10^{-5} and the normalized standard deviation is approximately 0.924575.

L. Direct CSI Baseline

Each normalized 100×62 window is flattened into a 6,200-dimensional vector. A balanced Random Forest with 300 trees is trained as a direct baseline.

M. Statistical Feature Representation

For each of 62 subcarriers, the notebook calculates mean, standard deviation, minimum, maximum, range, mean absolute temporal difference and standard deviation of temporal difference. This produces 434 features per window.

$$F = [\mu, \sigma, min, max, range, mean|\Delta x|, \sigma(\Delta x)]_{1 \dots 62}$$

N. Feature Importance and Ablation

Feature importance from the feature-based Random Forest shows several influential dimensions, including indices 124, 141, 125, 140 and 178. Ablation experiments under GroupKFold show that standard deviation is the strongest individual feature group in the principal SVM comparison.

O. Conventional Classifier Comparison

Model	Mean GroupKFold	Std. dev.
SVM RBF	0.708935	0.119149
Random Forest	0.691263	0.116319
HistGradientBoosting	0.654599	0.130405
KNN	0.653261	0.158341
Extra Trees	0.642927	0.176670
Logistic Regression	0.628302	0.132013
SVM Linear	0.595416	0.139859

Table 2. Conventional classifier comparison under recording-aware evaluation.

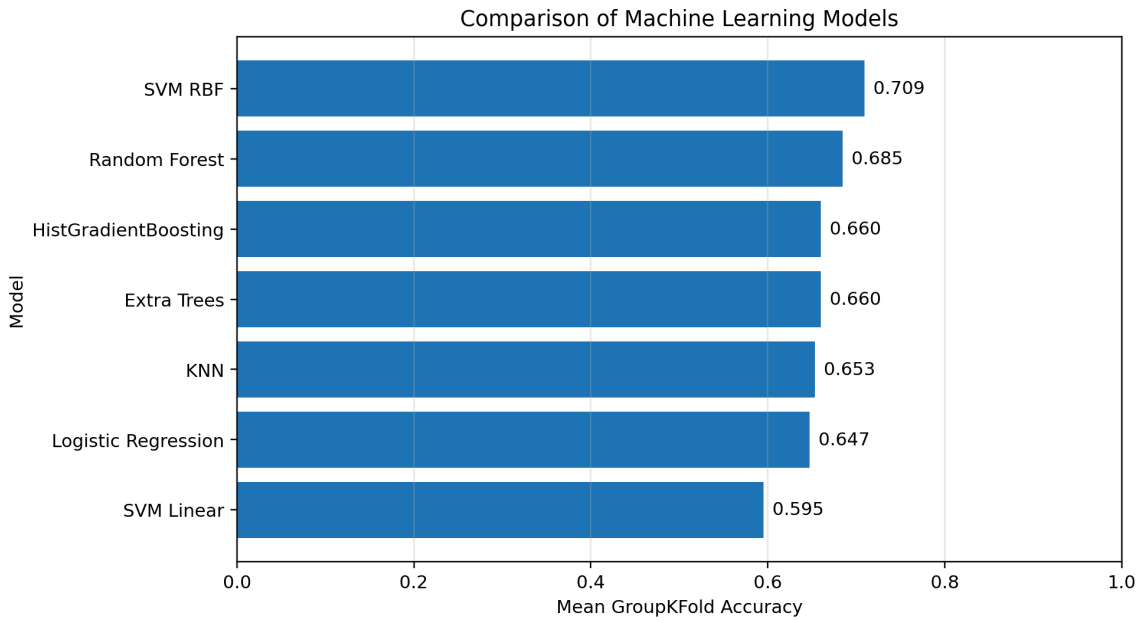


Figure 4. Original notebook plot: comparison of machine-learning models.

P. Feature-Group Selection

Feature set	Mean accuracy	Std. dev.
std	0.708935	0.119149
mean + std	0.703205	0.095588
std + mean_abs_diff	0.654924	0.151278
std + range	0.645454	0.119553
all features	0.644087	0.151437
mean	0.626793	0.151130
std + max	0.620677	0.183887
std + range + max	0.598486	0.147091
mean_abs_diff	0.557157	0.171454
range	0.552914	0.140778
std_diff	0.550592	0.170378
max	0.542075	0.169103
min	0.489360	0.234497

Table 3. Recording-aware performance for the principal SVM feature representations.

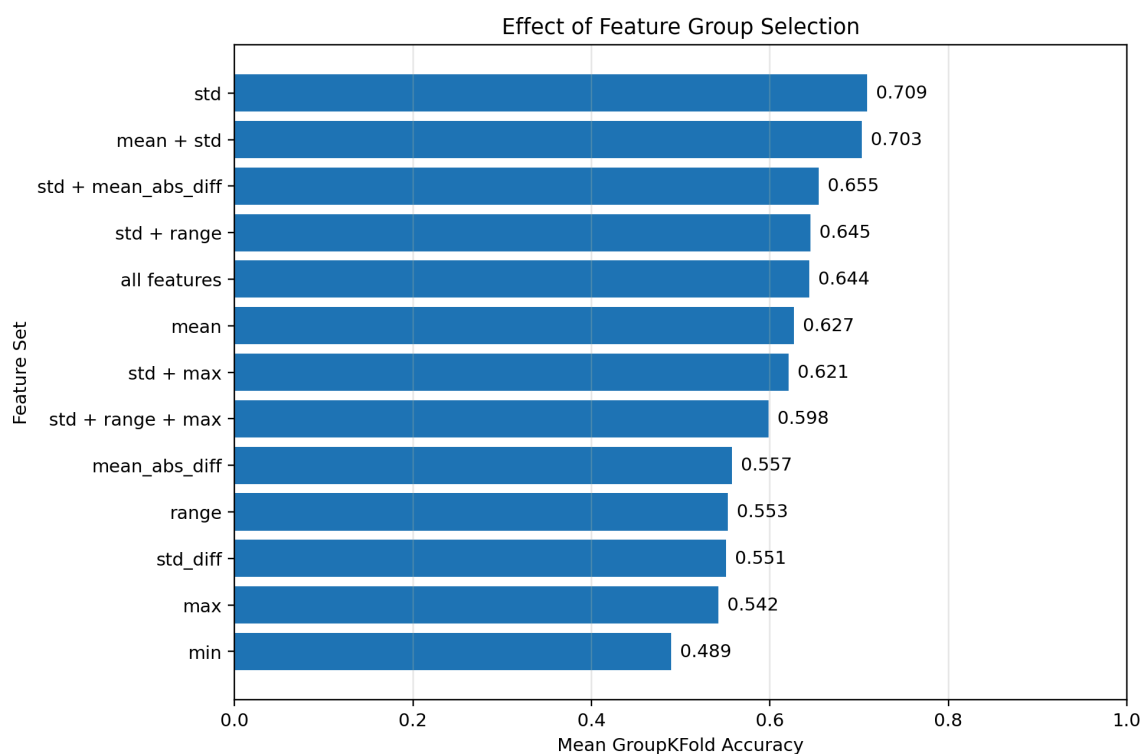


Figure 5. Original notebook plot: effect of feature-group selection.

Q. SVM Hyperparameter Search

C	Gamma	Mean accuracy	Std. dev.
100	0.001	0.711012	0.140379
1	scale	0.708935	0.119149
100	0.01	0.706466	0.114107
10	0.01	0.687928	0.178783
100	scale	0.684368	0.119721
1	0.01	0.680299	0.123592
10	0.001	0.660305	0.147284
10	scale	0.654391	0.174586
0.1	scale	0.603384	0.160904
1	0.001	0.598157	0.161392

Table 4. Leading SVM hyperparameter configurations.

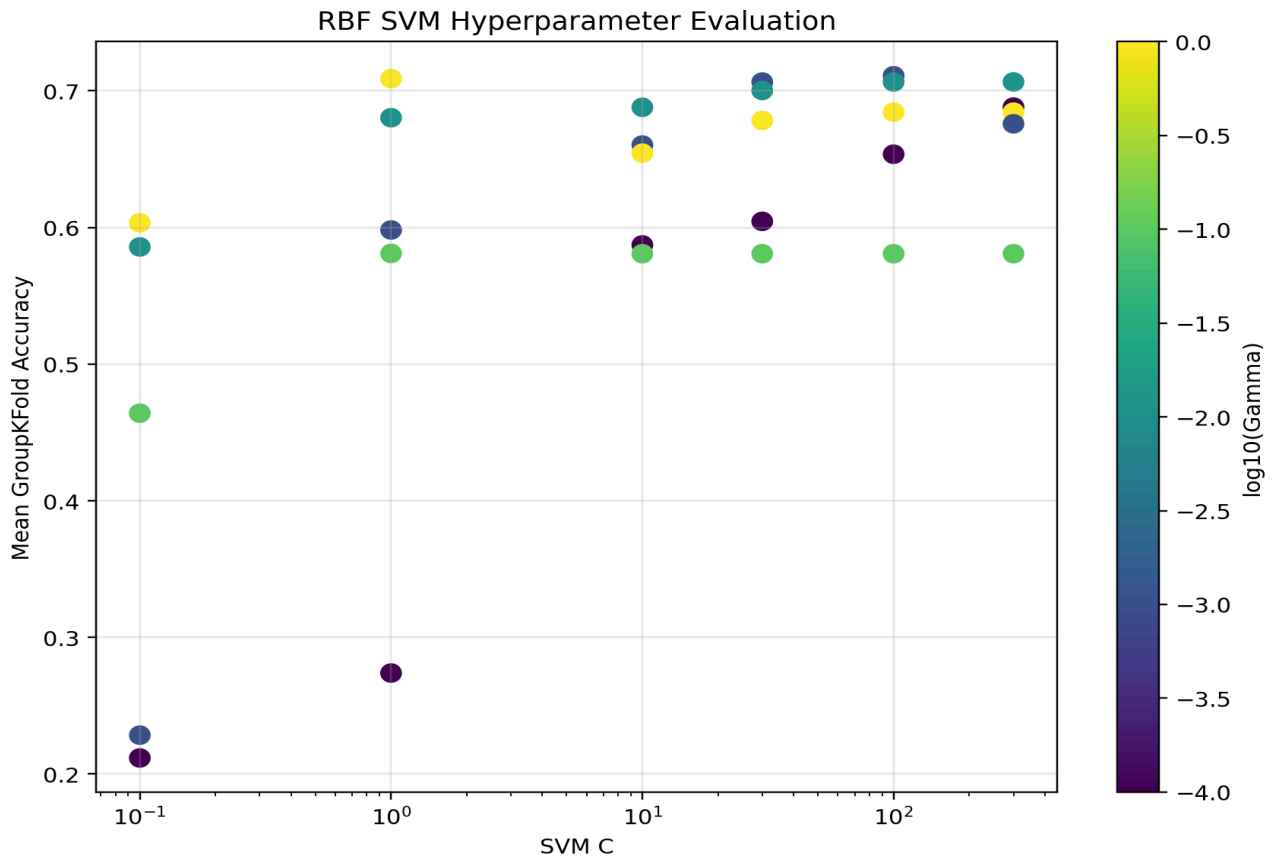


Figure 6. Original notebook plot: RBF-SVM hyperparameter evaluation.

R. PCA Dimensionality Reduction

Components	Mean accuracy	Std. dev.
10	0.722775	0.168633
25	0.716716	0.160043
30	0.711012	0.140379
40	0.711012	0.140379
50	0.711012	0.140379
20	0.699217	0.164567
15	0.693335	0.160494
5	0.660808	0.129797

Table 5. PCA dimensionality experiment.

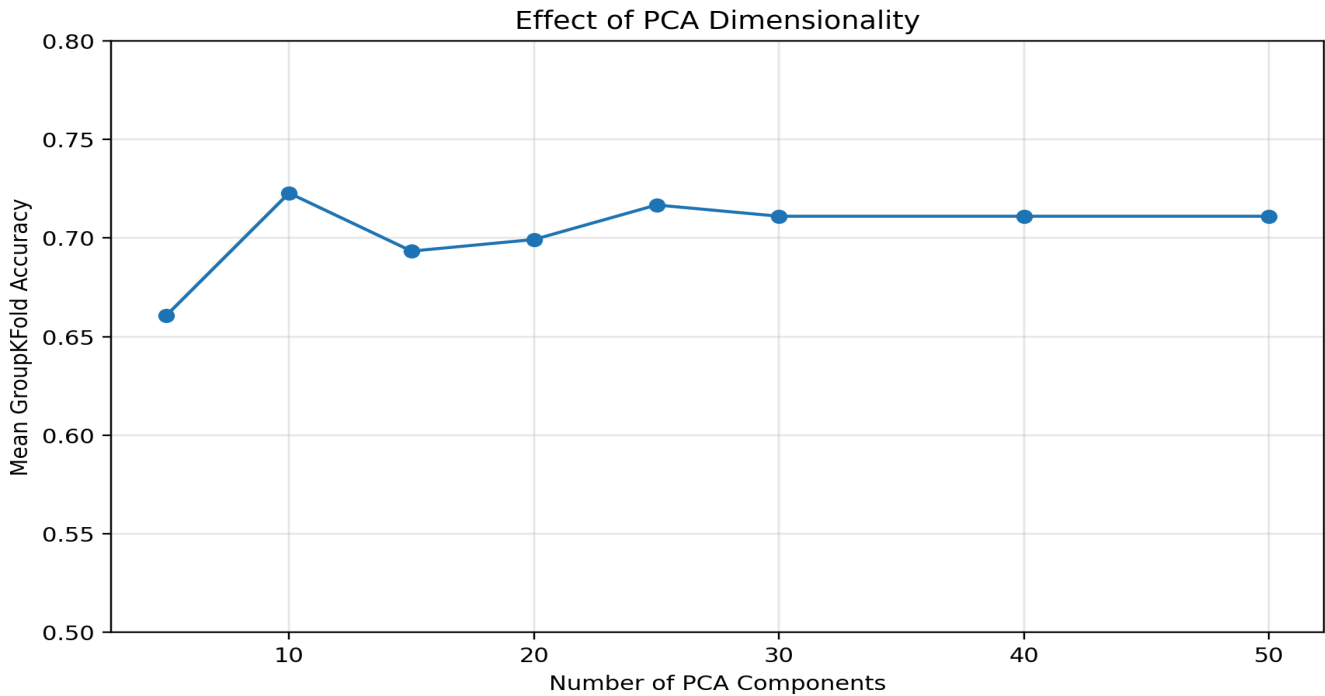


Figure 7. Original notebook plot: effect of PCA dimensionality.

S. Final Selected Model

The final pipeline is StandardScaler $\rightarrow$ PCA(10) $\rightarrow$ RBF-SVM, with $C = 100$ and $\gamma = 0.001$. It is retrained on all twelve training recordings before evaluation on the three untouched recording-05 files.

T. Reproducibility

The original notebook used an external dataset path and stored no output for many later cells. During preparation of this report, the same notebook logic was executed against the fifteen CSV recordings contained in the supplied csi_receiver archive. Where outputs were absent from the original notebook, numerical results were regenerated from the supplied dataset.

IV. Simulation Results & Discussions

A. Preliminary Simulation Results

The preliminary phase successfully deployed WiFi-DensePose in Docker and confirmed that the application could run in simulation mode without physical CSI hardware. The preliminary report explicitly states that these observations demonstrate software operation rather than sensing accuracy.

The simulation stage served as a software and conceptual bridge, allowing the authors to understand CSI acquisition, preprocessing, feature extraction, inference and downstream interfaces before working with the local recorded CSI dataset.

B. Dataset Quality Results

The raw dataset contains 12,737 packet rows. Sixteen rows fail the exact 130-value packet requirement and are removed. The retained 12,721 rows form the numerical basis of the activity-recognition experiments.

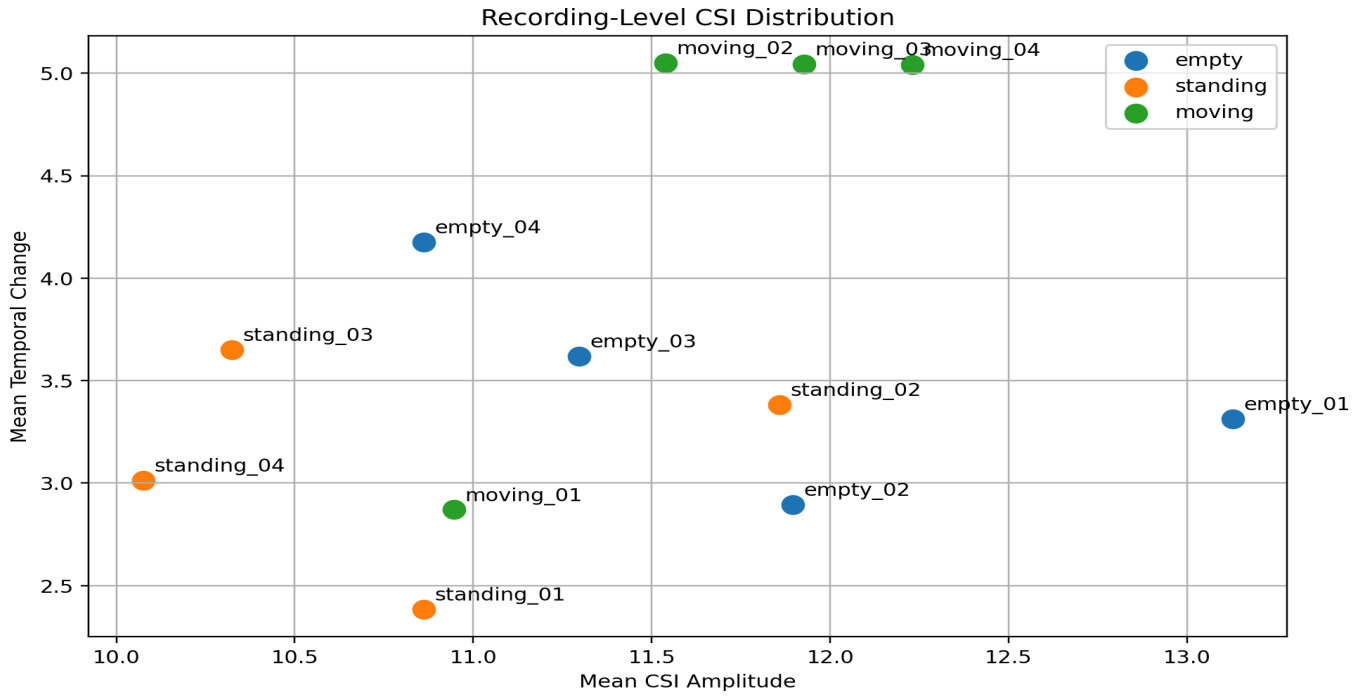


Figure 8. Original notebook plot: recording-level CSI feature distribution.

The recording-level statistics show noticeable variation even within a single class. Mean temporal change is generally higher for moving recordings, while amplitude statistics overlap between classes. This variation helps explain the importance of recording-aware validation.

C. Raw CSI and Amplitude Visualisation

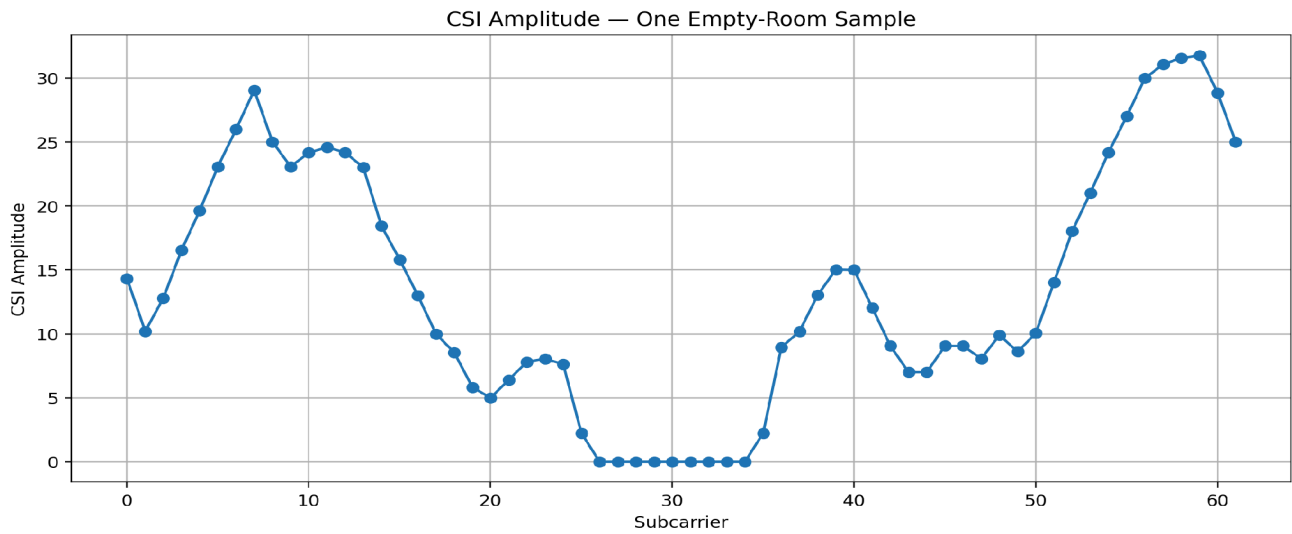


Figure 9. Original notebook plot: CSI amplitude across the 62 subcarriers for one empty-room sample.

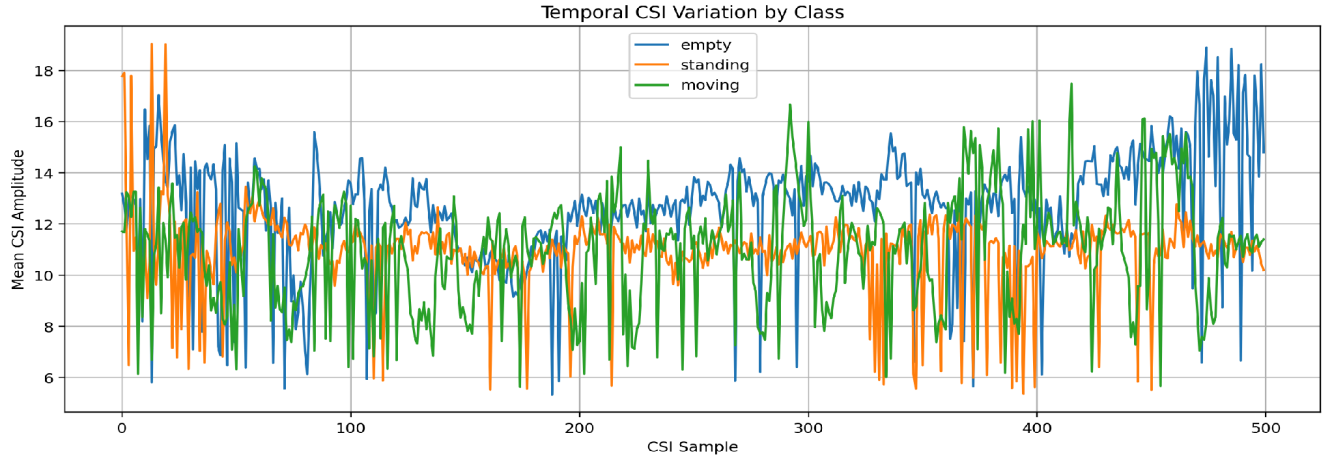


Figure 10. Original notebook plot: mean CSI amplitude trajectories for the three classes.

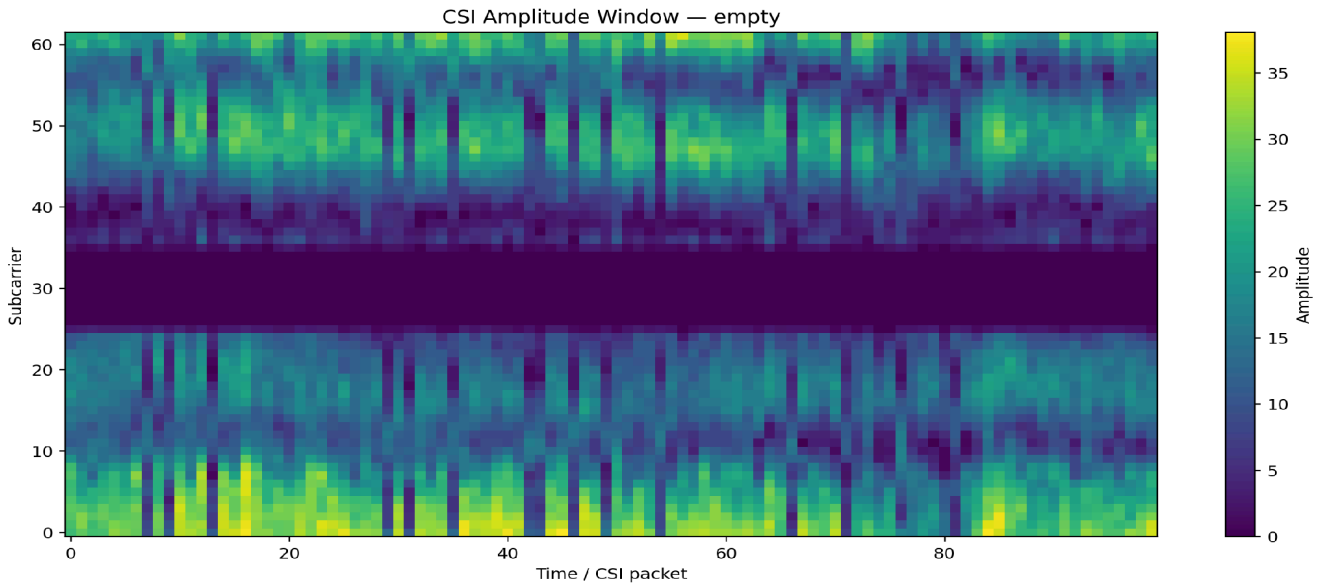


Figure 11. Original notebook plot: a 100×62 CSI amplitude window.

D. Direct CSI Random Forest Result

The direct flattened-window Random Forest reaches 0.649123 accuracy on the 57 unseen test windows. The classification report gives F1 scores of 0.62 for empty, 0.71 for moving and 0.60 for standing; moving has the highest recall at 0.89.

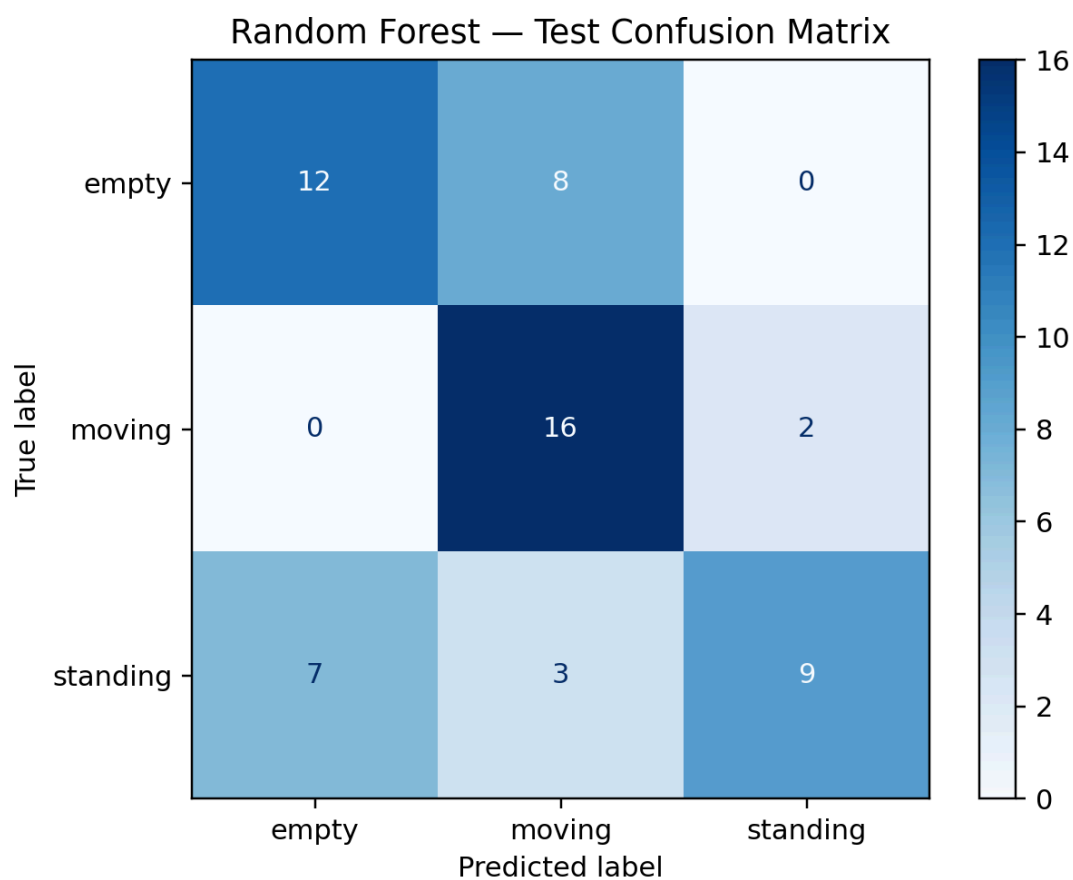


Figure 12. Original notebook plot: direct CSI Random Forest test confusion matrix.

E. Feature-Based Random Forest Result

The 434-feature Random Forest improves unseen-window accuracy to 0.912281. The moving class is perfectly classified in the held-out window set, while empty and standing achieve F1 scores of 0.88 and 0.86.

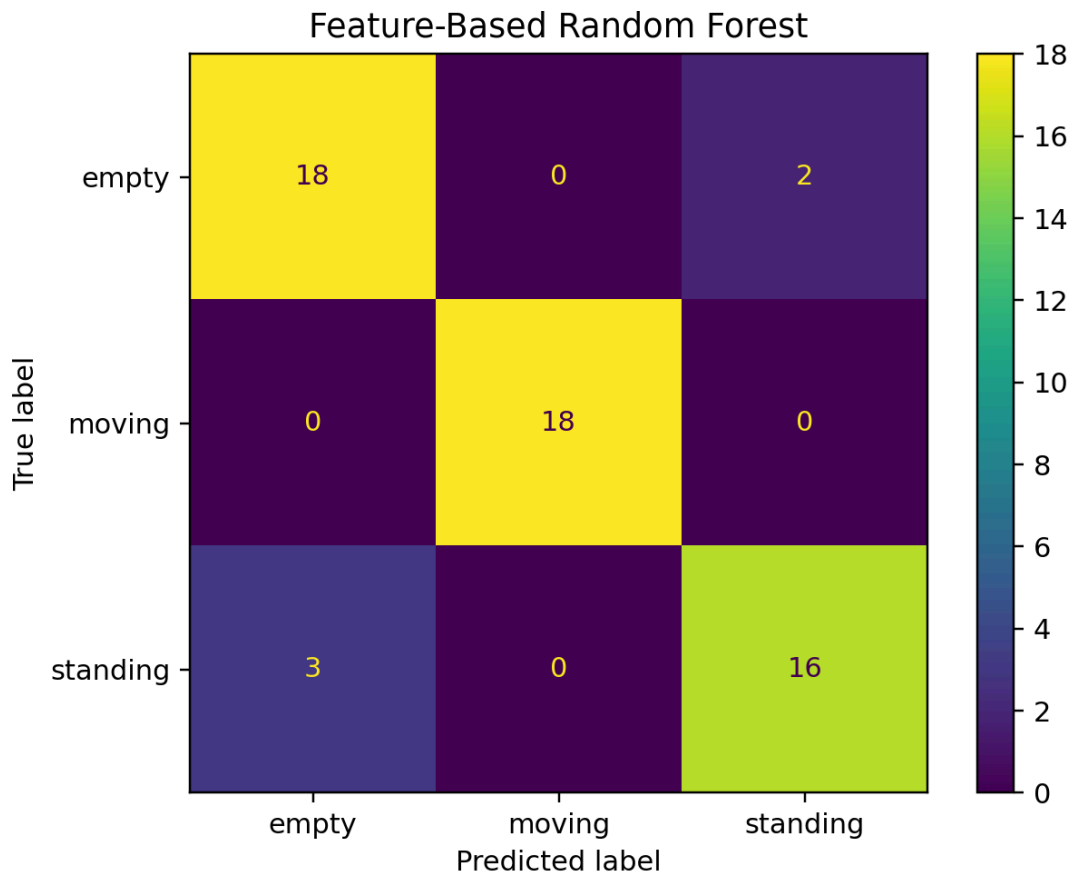


Figure 13. Original notebook plot: feature-based Random Forest confusion matrix.

F. Feature Importance

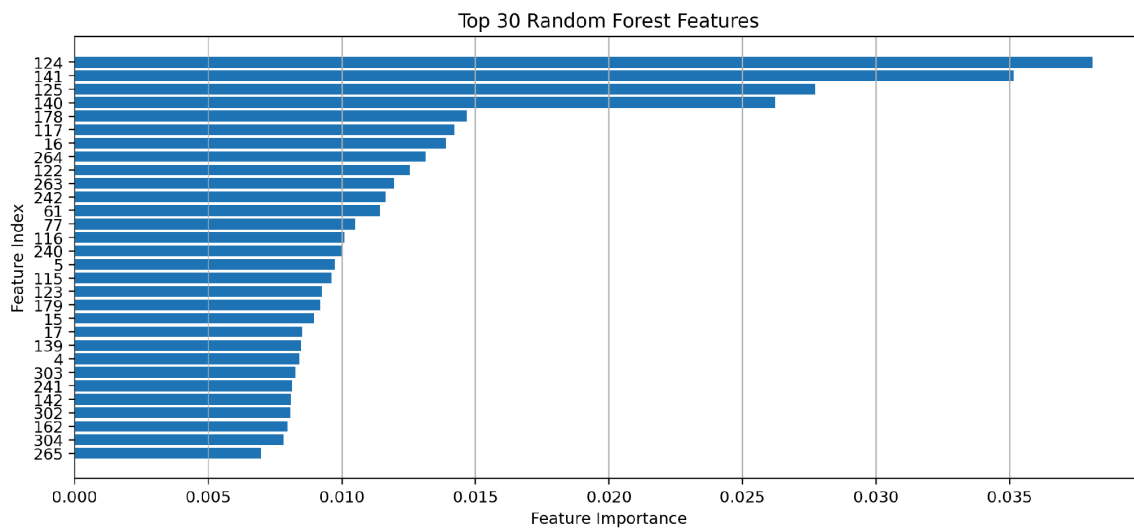


Figure 14. Original notebook plot: top 30 Random Forest feature importances.

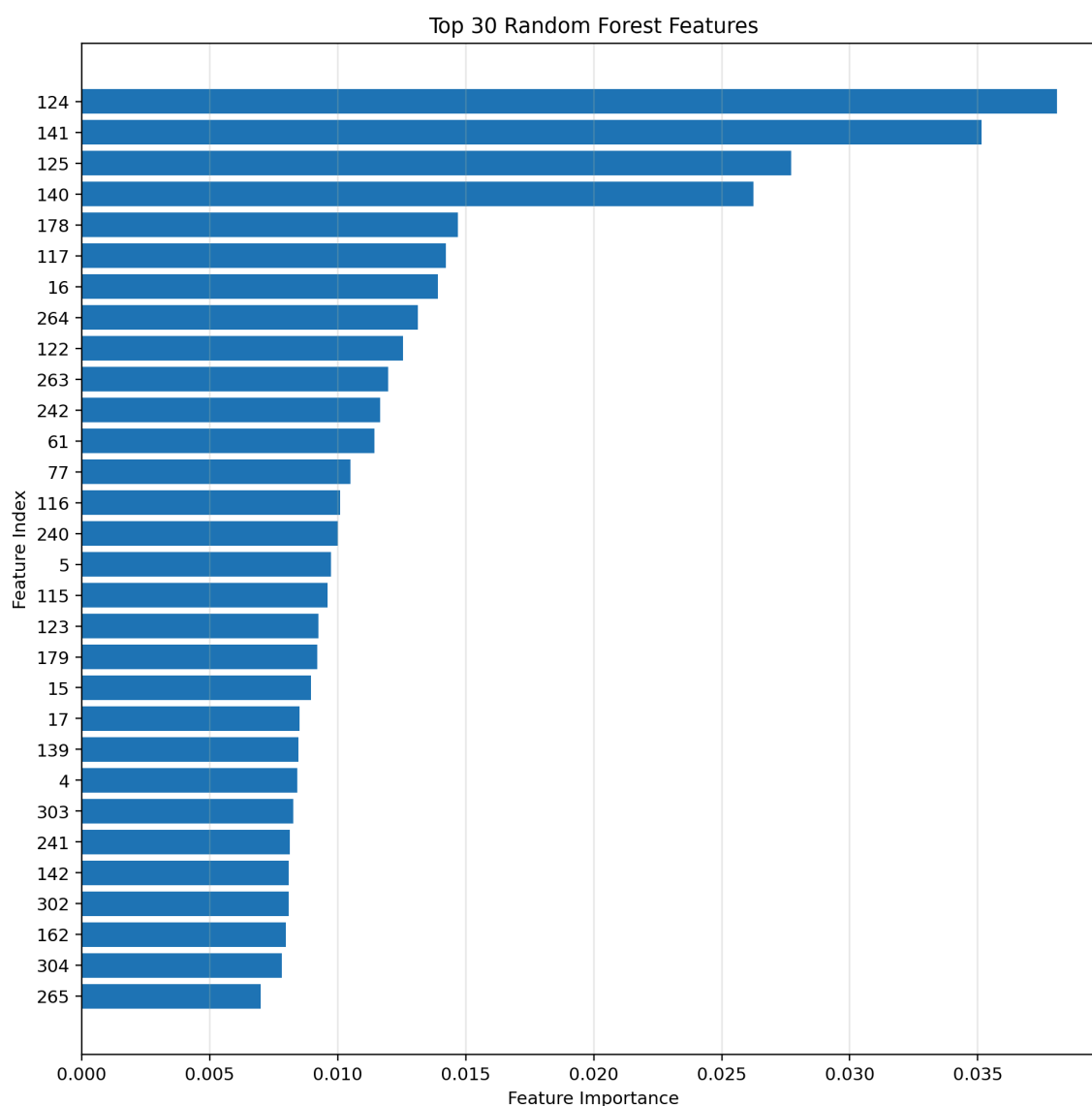


Figure 15. Original notebook plot: enlarged feature-importance view.

G. Stratified Versus Recording-Aware Validation

Five-fold stratified validation of the feature-based Random Forest gives 0.920000 mean accuracy with standard deviation 0.061012. When recording filenames are used as groups, the same representation gives 0.661165 mean accuracy with standard deviation 0.112066. This is a central result: random-window performance is not representative of performance on unseen recording conditions.

A detailed GroupKFold run reports fold accuracies of 0.666667, 0.852941, 0.594595, 0.676471 and 0.606061. A second recording-aware evaluation cell reports 0.679347 mean accuracy with standard deviation 0.103511 for its corresponding model stage.

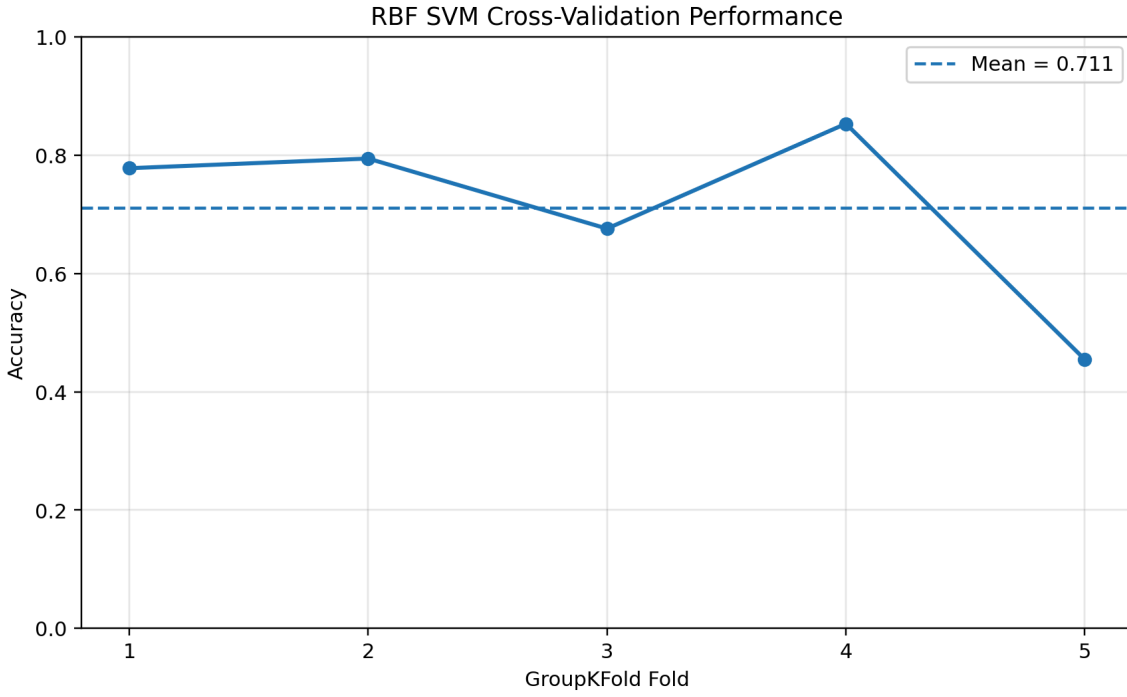


Figure 16. Original notebook plot: fold-wise RBF-SVM recording-aware cross-validation stability.

H. Model Comparison Under Recording-Aware Validation

RBF-SVM gives the highest mean accuracy in the principal comparison at 0.708935, followed by Random Forest at 0.691263. Linear SVM is lowest at 0.595416.

I. Feature-Group Analysis

Standard deviation is the strongest individual feature group in the principal SVM comparison. Mean plus standard deviation gives 0.703205, while all features give 0.644087. More features do not automatically improve recording-level generalisation.

J. Temporal-Only and STD+Range Experiments

Using only the 124 temporal-dynamics features gives 0.567942 mean recording-level accuracy with standard deviation 0.197571. The STD+Range representation gives 0.657931 mean accuracy with standard deviation 0.153649.

K. PCA-SVM Results

PCA with ten components gives the strongest recording-aware cross-validation result among the evaluated PCA sizes: 0.722775 mean accuracy with standard deviation 0.168633. The best SVM hyperparameters remain $C = 100$ and $\gamma = 0.001$.

L. Final Unseen Test Evaluation

The selected PCA-SVM is trained on all twelve training recordings and evaluated on the three recording-05 files. Window-level test accuracy is 0.578947. The class-wise results are: empty precision 0.42, recall 0.40, F1 0.41; moving precision 0.95, recall 1.00, F1 0.97; and standing precision 0.37, recall 0.37, F1 0.37.

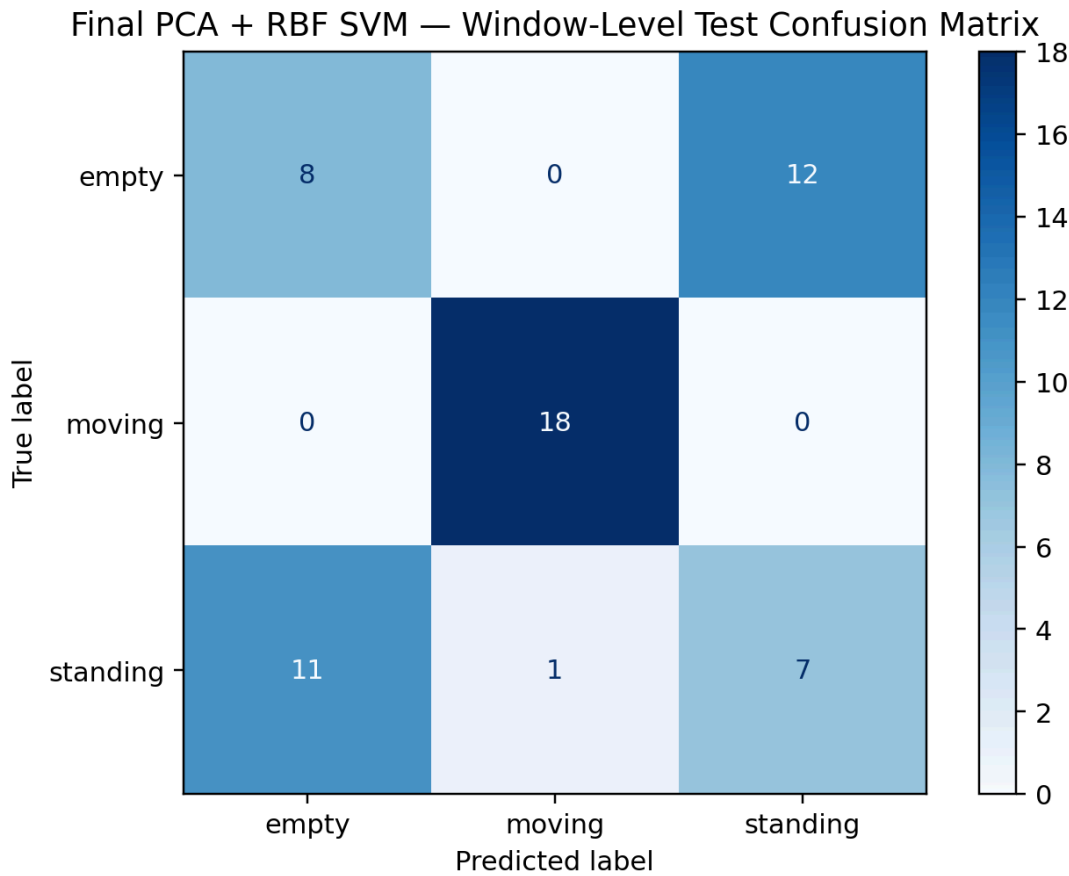


Figure 17. Original notebook plot: final PCA + RBF-SVM window-level confusion matrix.

M. Recording-Level Test Results

Recording	True	Windows	Window accuracy	Aggregated prediction	Correct
empty_05.csv	empty	20	0.400	standing	No
moving_05.csv	moving	18	1.000	moving	Yes
standing_05.csv	standing	19	0.368	empty	No

Table 6. Final unseen recording-wise performance of the selected PCA-SVM.

Majority voting gives 33.33% recording-level accuracy. Mean decision-score aggregation improves recording-level accuracy to 66.67%, correctly identifying empty_05 and moving_05 while misclassifying standing_05 as empty.

N. Confusion Matrix Analysis

The confusion matrices provide a more informative view of model behaviour than overall accuracy alone. At the final window level, the moving class is much better separated from the other two classes, while the main confusion occurs between empty and standing. This is consistent with the temporal behaviour observed in the CSI traces: movement repeatedly changes propagation paths, whereas a stationary person can produce comparatively stable CSI patterns that overlap with natural fluctuations in an empty environment.

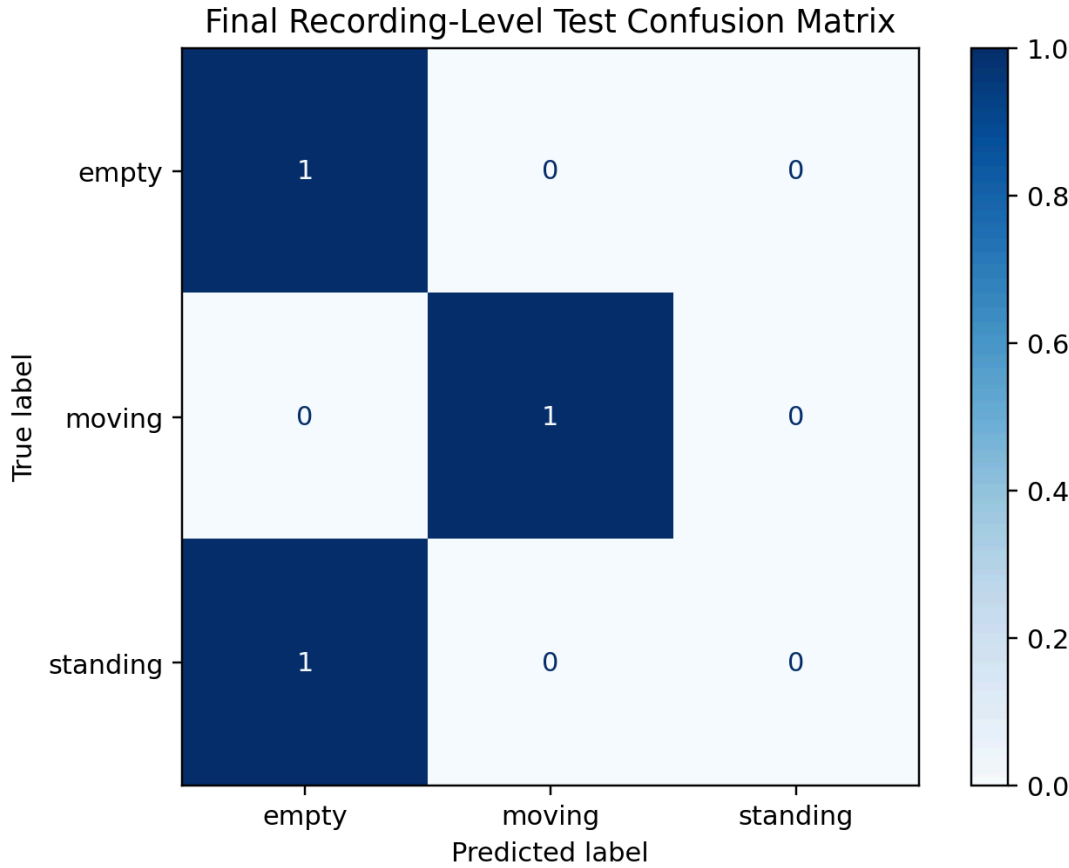


Figure 18. Original notebook plot: final recording-level confusion matrix.

O. Final Recording-Level Classification Report

Class	Precision	Recall	F1	Support
empty	0.50	1.00	0.67	1
moving	1.00	1.00	1.00	1
standing	0.00	0.00	0.00	1
Accuracy			0.67	3
Macro average	0.50	0.67	0.56	3
Weighted average	0.50	0.67	0.56	3

P. Discussion of Results

The strongest positive finding is that the CSI recordings contain enough information for coarse activity discrimination. The feature-based Random Forest reaches more than 90% accuracy on held-out windows. However, recording-aware validation is substantially lower and varies across folds.

Moving is consistently the easiest class to recognize. Movement produces stronger temporal changes in the CSI channel. Standing is more difficult because a stationary person can produce a channel state that resembles an empty environment. The final PCA-SVM misclassifies standing_05 as empty at the recording level.

The experiments also indicate strong session variation. The same activity can exhibit different amplitude and temporal-change statistics across recordings. This may arise from position, timing, environmental multipath, background activity, transmitter-receiver geometry or other uncontrolled factors.

Q. Relationship to WiFi-DensePose

R. Potential Applications

R.1 Smart Home and Occupancy Monitoring

A CSI-based system that distinguishes empty environments from broad human activity can provide an input to smart-home automation, including occupancy-aware lighting, ventilation and energy-management decisions.

R.2 Elderly and Assisted-Living Monitoring

Camera-free sensing could support general occupancy and movement monitoring in assisted-living environments. Because such applications can become safety-sensitive, substantially larger datasets and independent validation would be necessary before deployment.

R.3 Smart Buildings

Wireless occupancy information can contribute to room-utilisation and building-management systems, where activity information may be combined with lighting, heating, ventilation and air-conditioning control.

R.4 Privacy-Preserving Human Sensing

CSI sensing does not directly produce conventional visual imagery, making it potentially attractive where video monitoring is undesirable. Practical systems would still require explicit privacy, security and consent mechanisms.

R.5 Industrial Monitoring

CSI can potentially provide broad occupancy or movement detection in designated industrial areas. With additional classes and diverse recordings, the current pipeline could be extended to different workplace states.

The present classifier is a sensing baseline rather than a pose-estimation system. Person-in-WiFi [3] demonstrated fine-grained person perception from WiFi, while DensePose From WiFi [2] investigated dense human body correspondence from WiFi amplitude and phase. The current project establishes an intermediate capability: coarse activity recognition from locally recorded CSI while exposing the generalisation issues that must be solved before fine-grained pose estimation.

S. Limitations

The principal limitation is the scale and diversity of the current dataset. It contains only fifteen recordings and three activity classes, with only one held-out recording per class in the final test. This means that recording-level test statistics must be interpreted cautiously. The available recordings also do not by themselves establish subject-independent or environment-independent generalisation, because the report can only evaluate variation represented in the supplied data.

A second limitation concerns the sensing representation and supervision. The baseline primarily uses CSI amplitude and does not implement the complete phase-sanitization pipeline discussed in the preliminary WiFi-DensePose study. The present recordings also lack synchronized RGB/DensePose ground truth, so the experiment cannot train or validate a true CSI-to-pose model. Finally, overlapping temporal windows are correlated, which makes recording-aware grouping essential; the current work addresses this issue in validation but does not remove the underlying dependence in the dataset.

T. Future Work

The first priority for future development is dataset expansion. The next dataset should include more subjects, more recording sessions, multiple transmitter-receiver placements and multiple room geometries. Ideally, CSI recordings should be synchronized with RGB video, DensePose or keypoint annotations, and environmental metadata so that wireless observations can later be mapped to a spatial body representation rather than only an activity label.

The second priority is richer signal processing. Future experiments should investigate phase sanitization, amplitude-phase fusion, subcarrier selection, frequency-domain and time-frequency representations, wavelets and other features that preserve motion and spatial information. These experiments should be evaluated with subject-independent and environment-independent splits so that improvement reflects genuine robustness rather than similarity between recording sessions.

The third priority is model development. The current conventional-ML baseline provides a reference point for evaluating more capable architectures such as one-dimensional or two-dimensional convolutional networks, recurrent networks and transformer-based temporal models. Transfer learning and domain adaptation are also promising directions for reducing the mismatch between different wireless environments and for eventually adapting CSI representations to the WiFi-DensePose framework.

The long-term objective is to progress from coarse activity recognition to finer posture and pose estimation. This will

require richer labels, synchronized visual ground truth, more representative data and models capable of learning spatial body structure. Real-time inference, computational efficiency and embedded deployment should also be studied as the system moves from laboratory experimentation toward practical WiFi sensing applications.

T.1 Dataset Expansion

The next dataset should include more subjects, additional sessions, different transmitter-receiver placements, multiple room geometries, different distances, subject positions, movement speeds and directions, furniture arrangements and varying levels of wireless interference. A larger dataset would also allow more independent recordings to be reserved for final testing.

T.2 Synchronized Ground Truth

For progression toward WiFi-DensePose, CSI recordings should ideally be synchronized with RGB video, DensePose annotations or human keypoints. Environmental metadata should also be recorded so that wireless observations can later be related to spatial body information rather than only activity labels.

T.3 Richer CSI Processing

Future experiments should investigate phase sanitization, amplitude-phase fusion, subcarrier selection, frequency-domain representations, time-frequency analysis, wavelets and additional temporal or spectral features. Subject-independent and environment-independent evaluation should accompany these changes.

T.4 Additional Activity Classes

The present three-class problem can be expanded to include sitting, walking, lying down, entering a room, leaving a room and other controlled movements. A broader activity vocabulary would make the baseline more representative of practical human activity recognition.

T.5 Deep Learning-Based CSI Representation

The conventional models used here provide a reference point for deeper architectures. Future work can investigate 1-D and 2-D convolutional networks, recurrent models and transformer-based temporal models capable of learning richer representations directly from CSI sequences.

T.6 Domain Adaptation

The observed cross-recording variation suggests that practical deployment will require adaptation to changes in room geometry, device placement and subject identity. Transfer learning, domain-adversarial approaches and calibration procedures can therefore be investigated.

T.7 Human Pose Estimation

The longer-term objective is to move from coarse activity recognition toward human pose estimation. This requires richer supervision, synchronized visual ground truth, larger and more diverse CSI datasets and models capable of learning spatial relationships associated with human body structure.

T.8 Real-Time Implementation

The present analysis relies on recorded CSV files and notebook-based processing. A practical extension would integrate acquisition, packet validation, window generation, feature extraction, classification and decision aggregation into a continuous real-time pipeline.

T.9 Multiple Sensing Nodes

Multiple ESP32 sensing nodes can be investigated to provide spatially distributed wireless links. Complementary observations may improve robustness to changes in subject position and orientation and could eventually support richer spatial inference.

U. Project Contribution and Novelty Positioning

The present work should not be described as a new CSI classification algorithm. CSI-based activity recognition,

handcrafted feature extraction and conventional classifiers are established research directions. The contribution of this internship is the construction and analysis of a recording-aware baseline connected directly to the WiFi-DensePose research trajectory. The project documents the transition from preliminary simulation to real recorded CSI processing, compares multiple classical models and exposes the generalisation gap between random-window validation and unseen recordings.

V. Conclusions

A. Summary of the Work

B. Final Experimental Findings

C. Overall Conclusion

The final model was retrained on the twelve development recordings and tested on the three untouched recording-05 files. Window-level accuracy was 57.89%; moving was the most reliably recognized class, while standing remained difficult. At the recording level, mean decision-score aggregation correctly identified two of the three unseen recordings, giving 66.67% accuracy. Since the final test contains only three independent recordings, this result is best treated as an experimental observation rather than a statistically strong estimate of deployment performance.

The feature-based Random Forest achieved 91.23% accuracy on held-out test windows, while recording-aware validation was substantially lower. The RBF-SVM produced the strongest result in the principal conventional classifier comparison, and PCA with ten components produced a mean recording-aware accuracy of 72.28%. The selected final pipeline consisted of StandardScaler, PCA(10), and an RBF-SVM with $C = 100$ and $\gamma = 0.001$.

The internship progressed from a preliminary technical investigation of WiFi-DensePose to an experimentally evaluated CSI-based human activity-recognition pipeline. The final dataset contains fifteen recordings across empty, moving and standing classes. After packet validation, 12,721 valid CSI packets were retained from 12,737 raw rows. Each valid packet was converted to 62 subcarrier amplitudes derived from 62 I/Q pairs, and the sequences were segmented into 100-packet windows with a 50-packet step.

The internship progressed from a preliminary technical study of WiFi-DensePose to an experimentally verified CSI activity-recognition pipeline.

The dataset contains fifteen recordings across empty, standing and moving classes. After packet validation, 12,721 CSI packets were retained from 12,737 raw rows. Each valid packet was represented using 62 subcarrier amplitudes derived from 62 I/Q pairs.

The experiments show that statistical feature engineering is useful. A 434-feature Random Forest obtains 91.23% accuracy on unseen windows. However, recording-aware validation is substantially lower. An RBF-SVM provides the strongest conventional classifier in the principal comparison, and PCA with ten components raises its recording-aware cross-validation accuracy to 72.28%.

The final untouched test gives 57.89% window-level accuracy. Mean decision-score aggregation gives 66.67% recording-level accuracy, with moving_05 and empty_05 correctly identified and standing_05 misclassified as empty.

The central conclusion is not that the project has solved WiFi-based human pose estimation, but that it has established a reproducible first-stage sensing baseline and identified cross-recording generalisation as a major obstacle. This is directly relevant to the longer-term WiFi-DensePose objective.

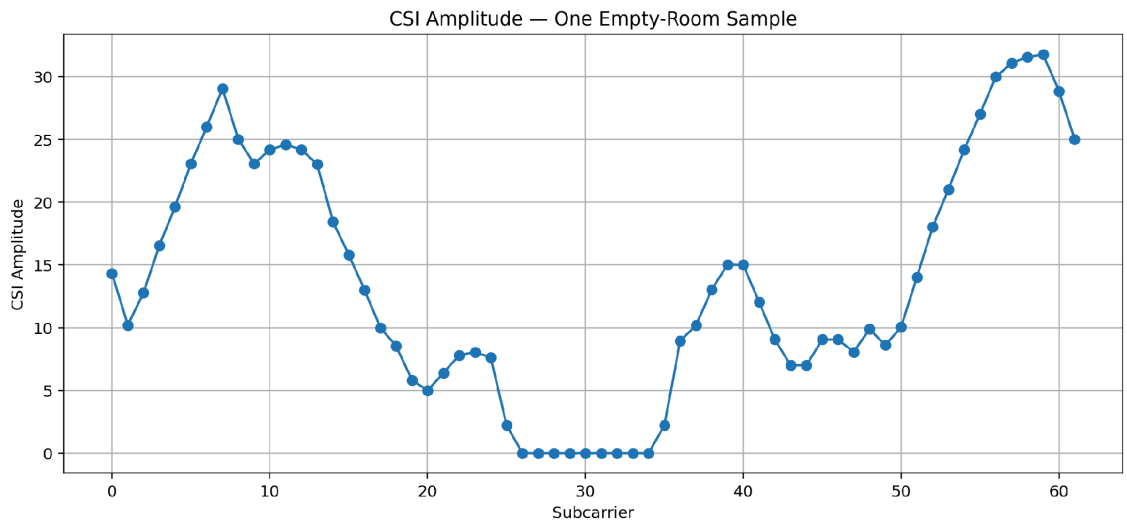
Future work should focus on larger synchronized CSI/visual datasets, phase-aware processing, environment- and subject-independent evaluation, deep sequence modelling and eventual CSI-to-pose representation learning.

VI. References

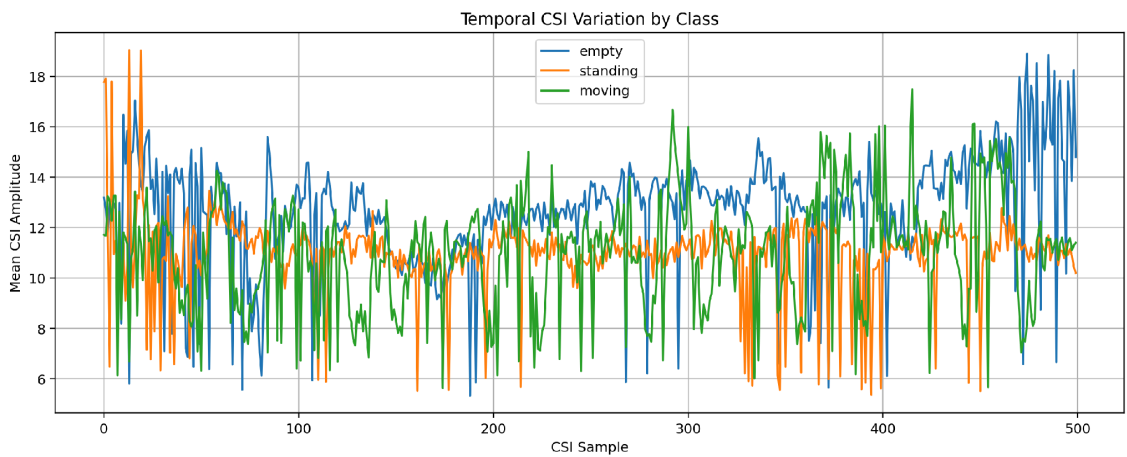
- [1] D. Akpele, “WiFi DensePose: Production-ready implementation of InvisPose,” GitHub repository. Available at: <https://github.com/davidakpele/wifi-densepose>.
- [2] J. Geng, D. Huang, and F. De la Torre, “DensePose From WiFi,” arXiv:2301.00250, 2022. Available at: <https://arxiv.org/abs/2301.00250>.
- [3] F. Wang, S. Zhou, S. Panev, J. Han, and D. Huang, “Person-in-WiFi: Fine-Grained Person Perception Using WiFi,” in Proceedings of the

Appendix B. Experimental Results and Notebook Outputs

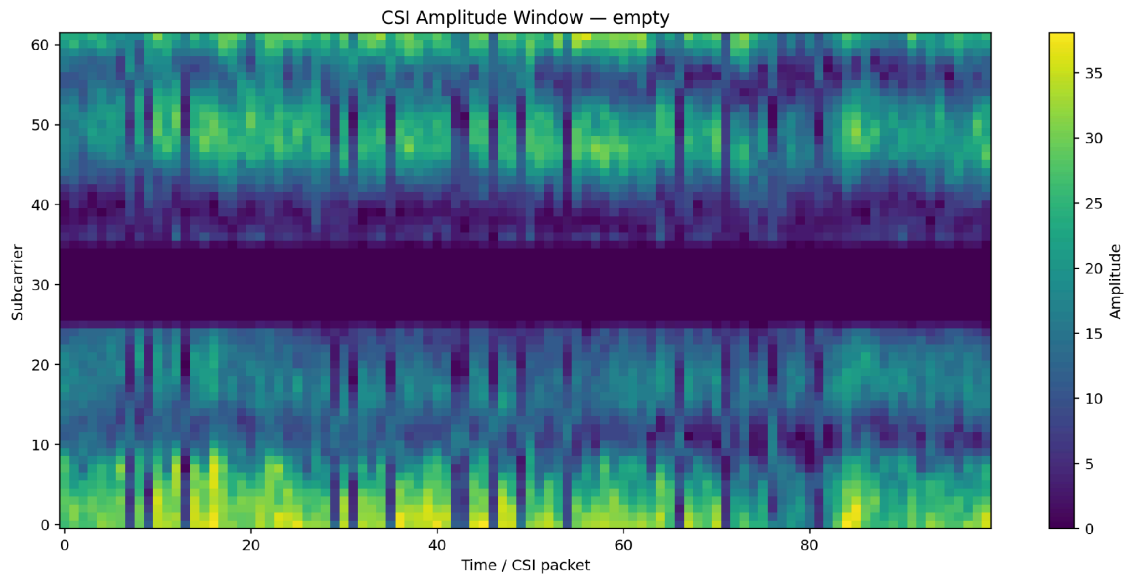
The following figures reproduce the principal plots generated by the supplied CSI_Activity_Recognition.ipynb after executing the notebook logic against the supplied dataset.



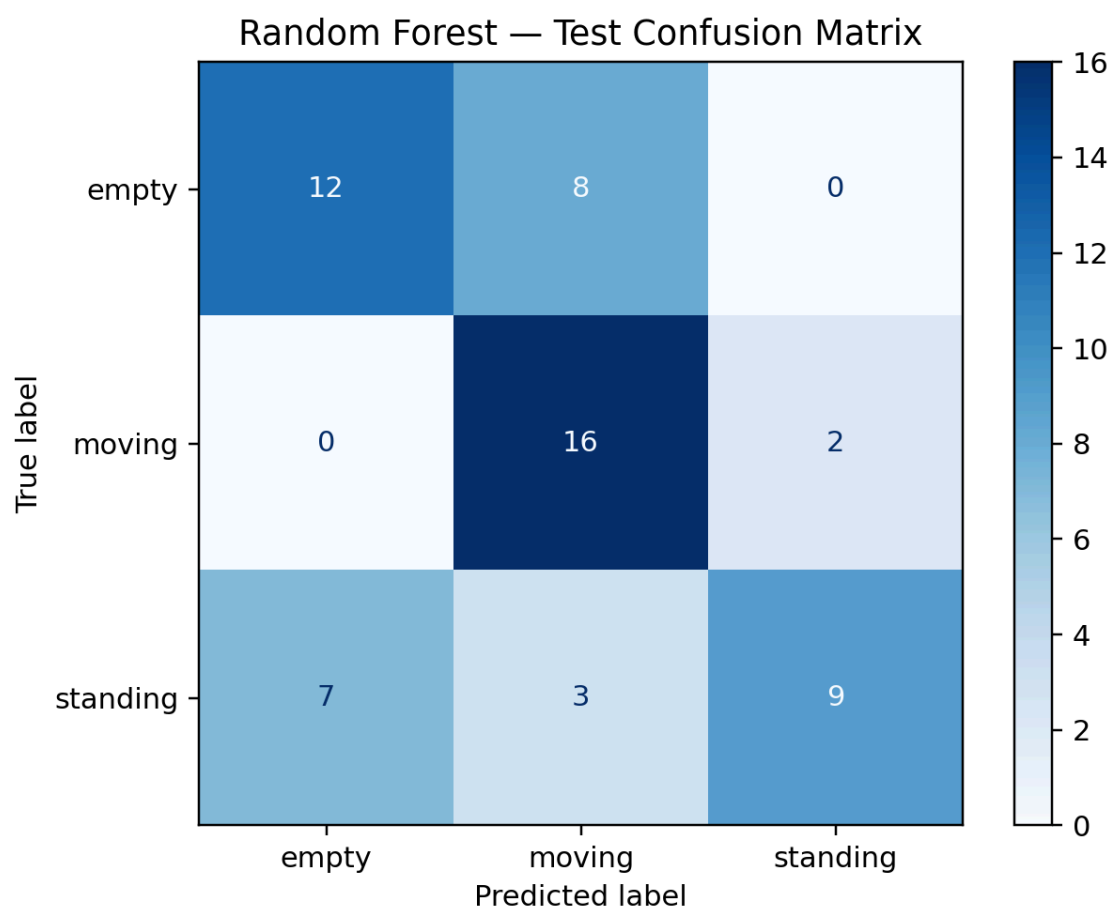
B.1 CSI amplitude for one empty-room sample.



B.2 Mean CSI amplitude trajectories for the three classes.



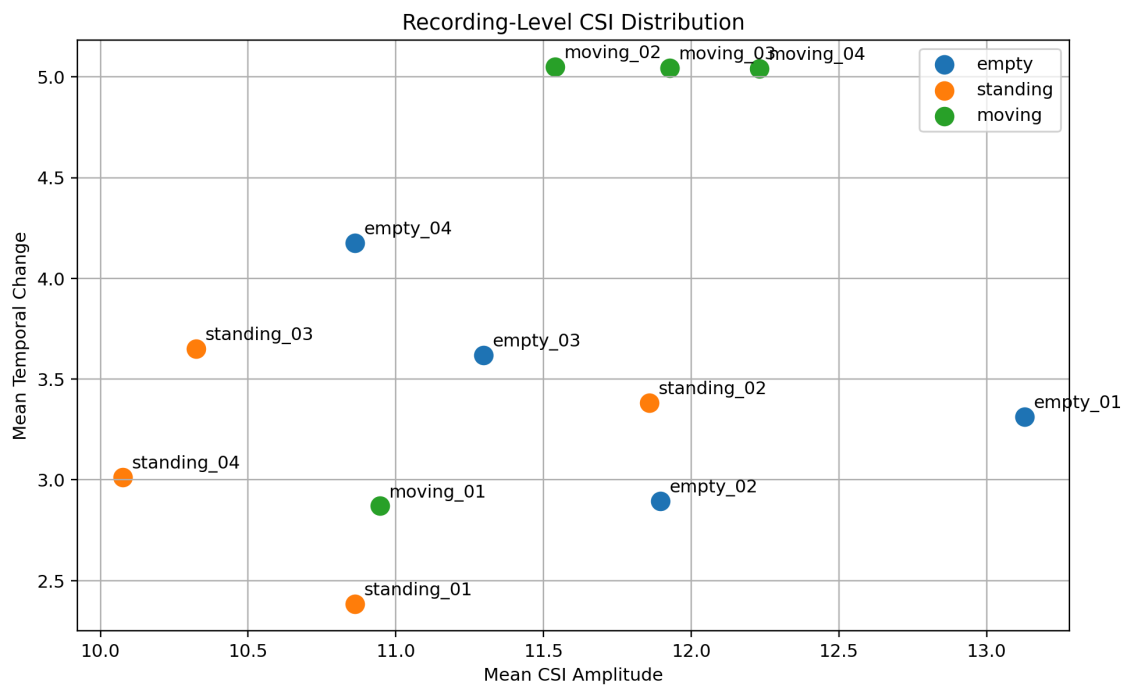
B.3 CSI amplitude window heatmap.



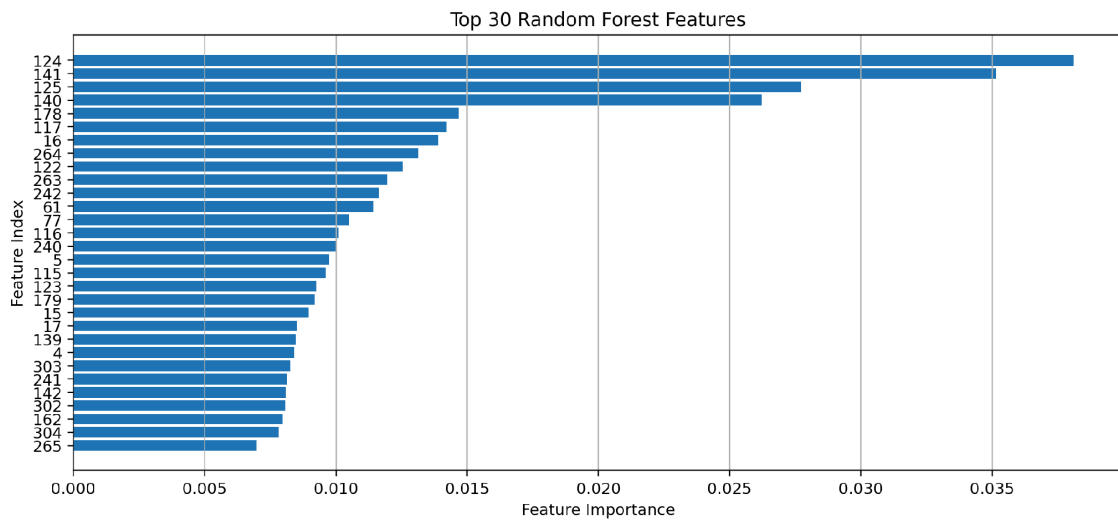
B.4 Direct CSI Random Forest confusion matrix.



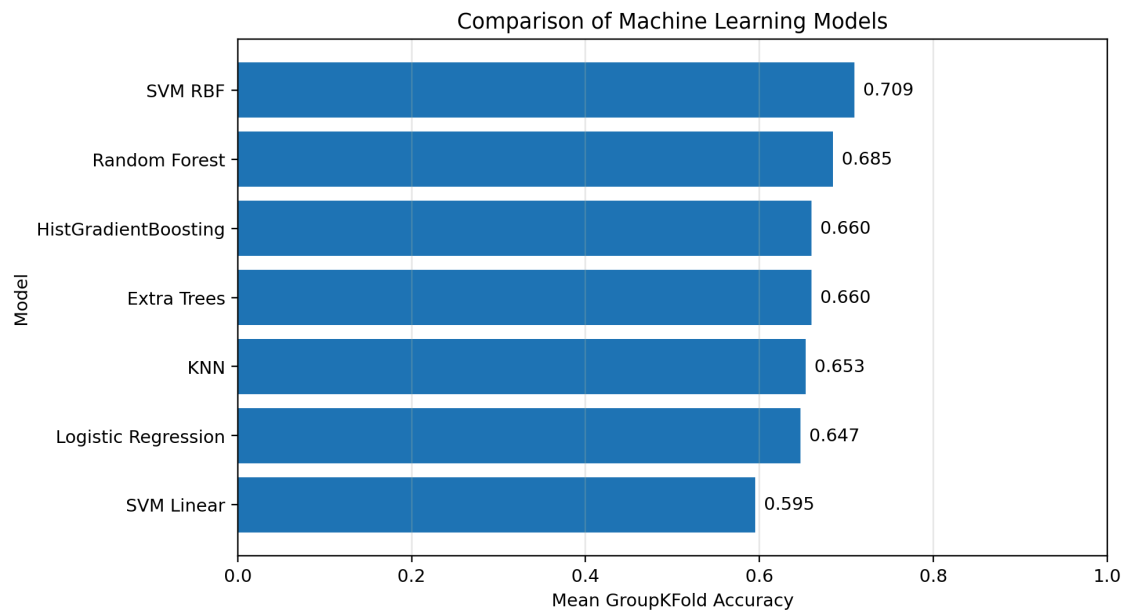
B.5 Feature-based Random Forest confusion matrix.



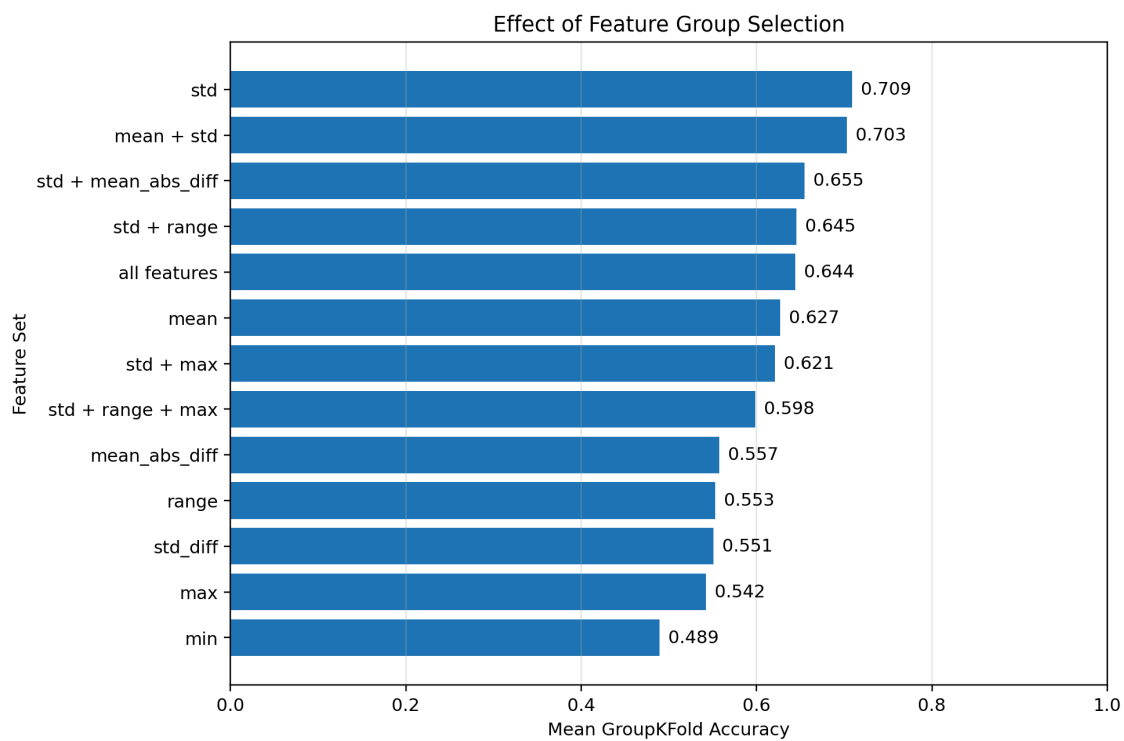
B.6 Recording-level CSI feature distribution.



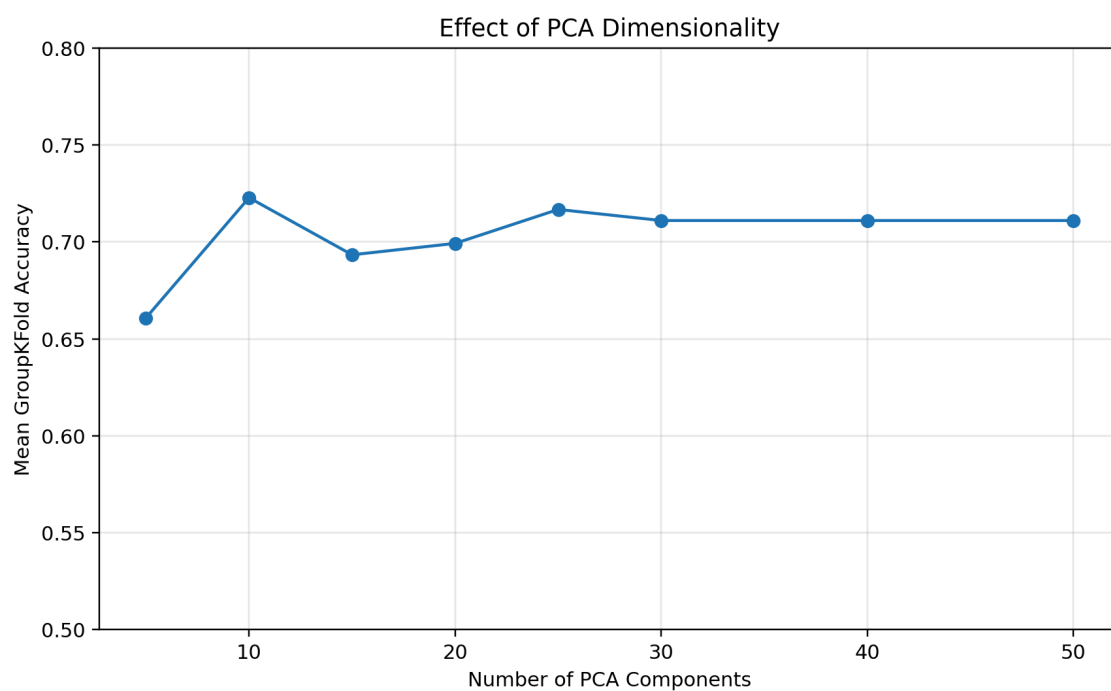
B.7 Top Random Forest feature importances.



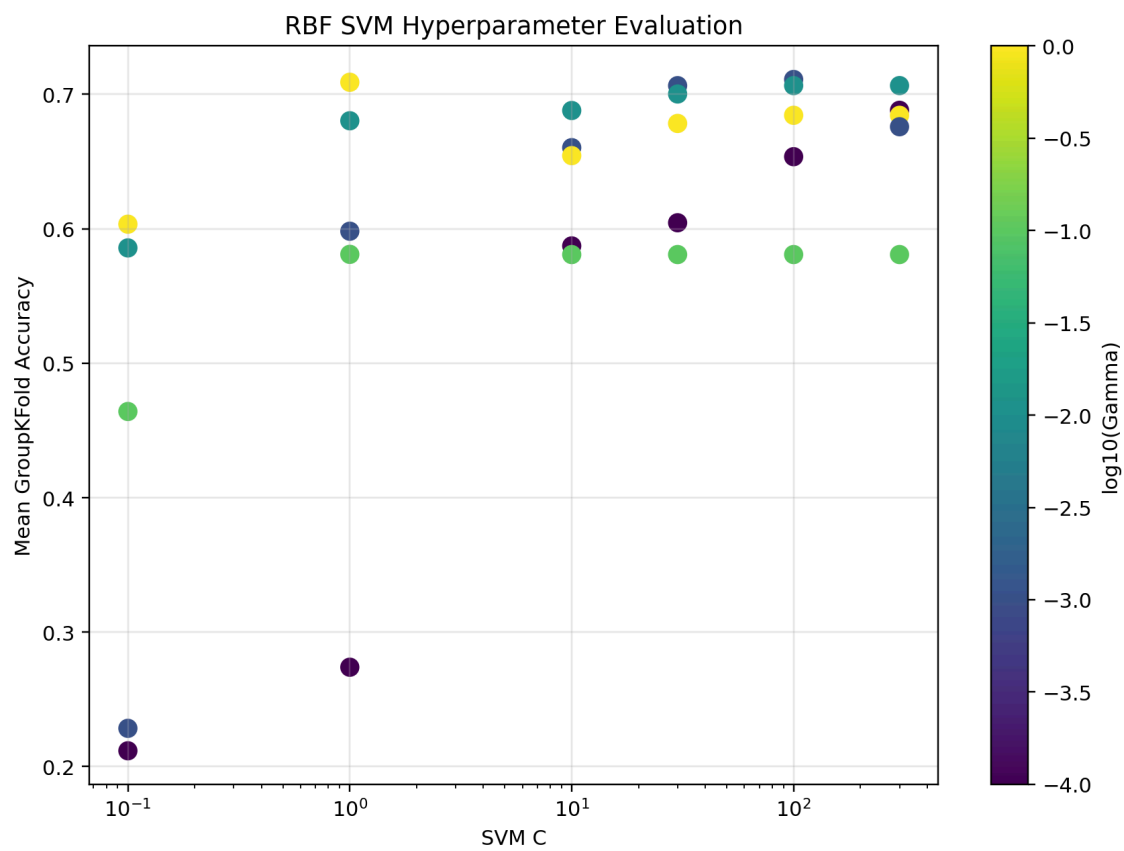
B.8 Model comparison under GroupKFold.



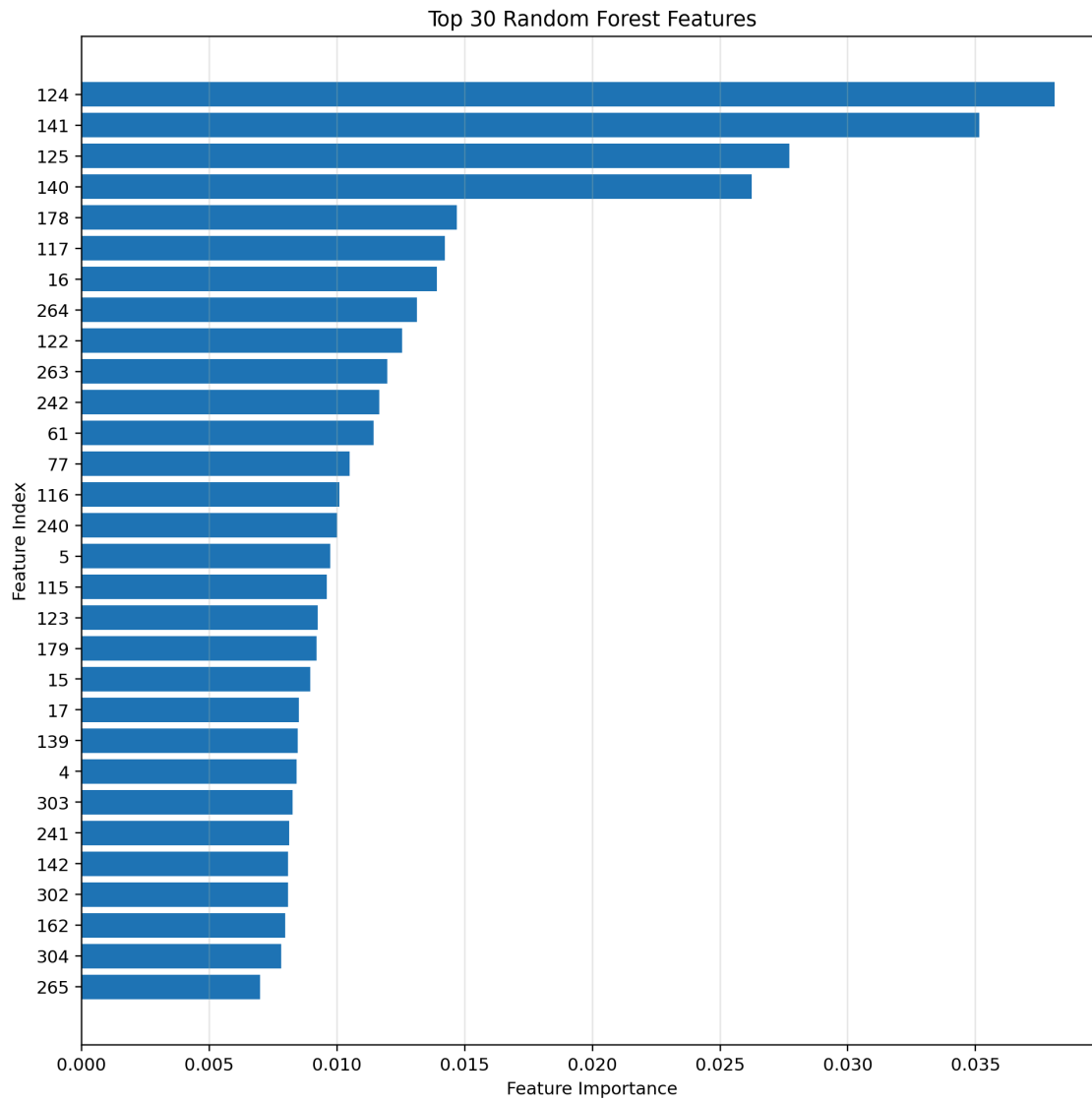
B.9 Feature-set comparison under GroupKFold.



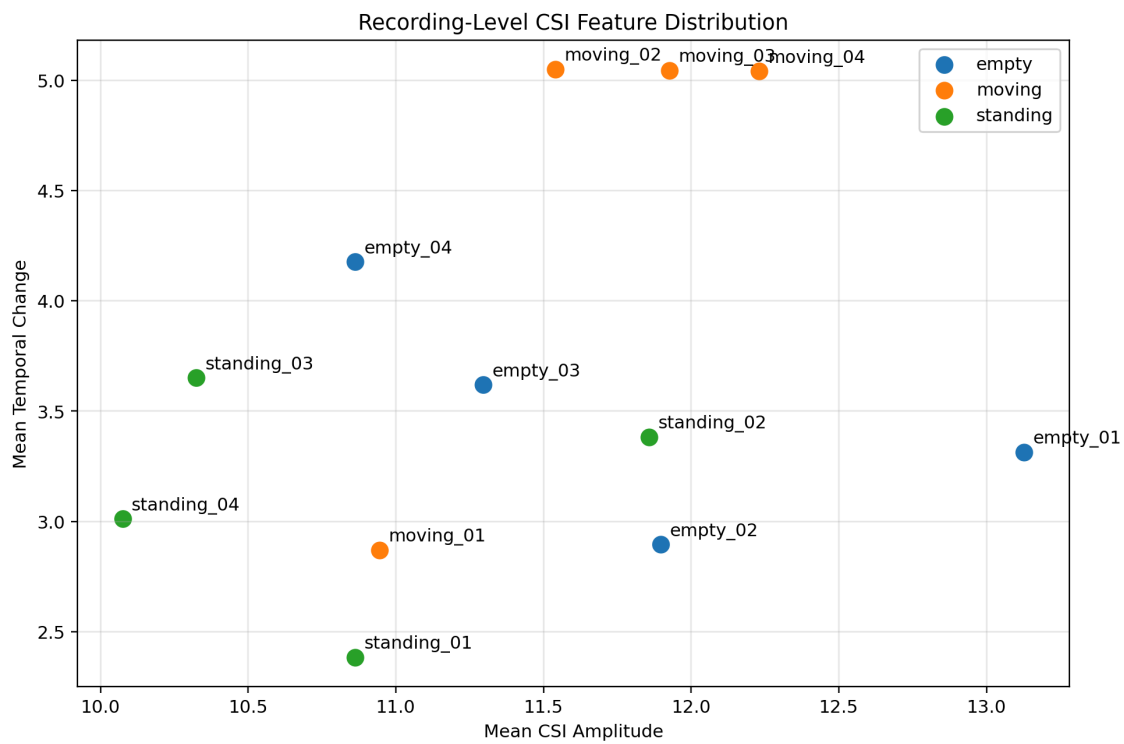
B.10 PCA dimensionality experiment.



B.11 SVM hyperparameter evaluation.

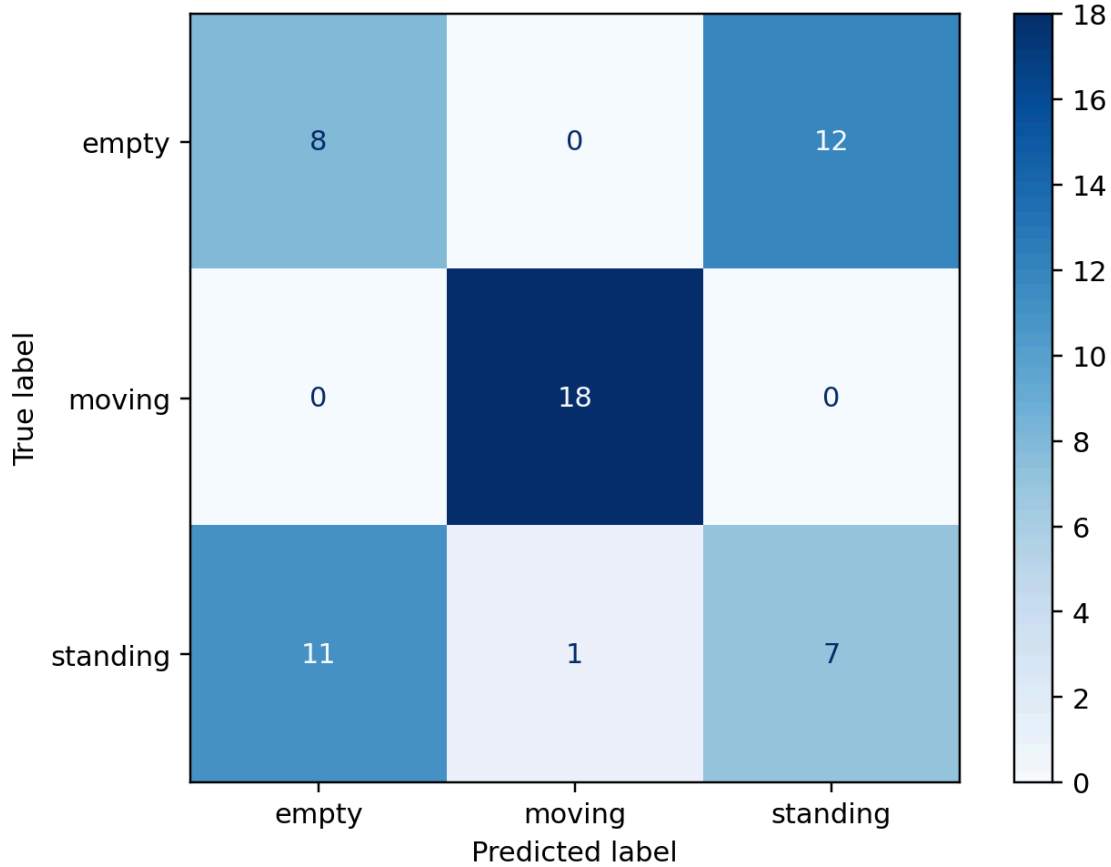


B.12 Expanded feature-importance plot.

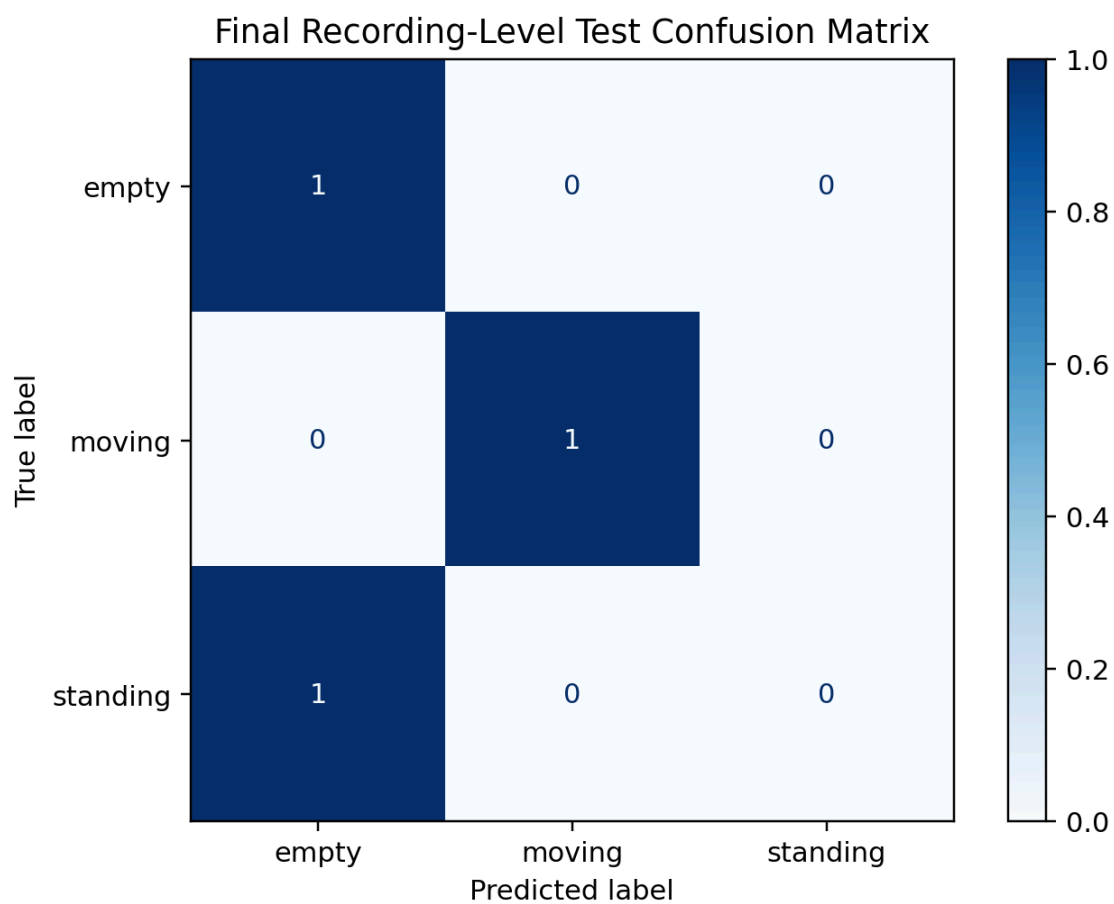


B.13 CSI recording distribution.

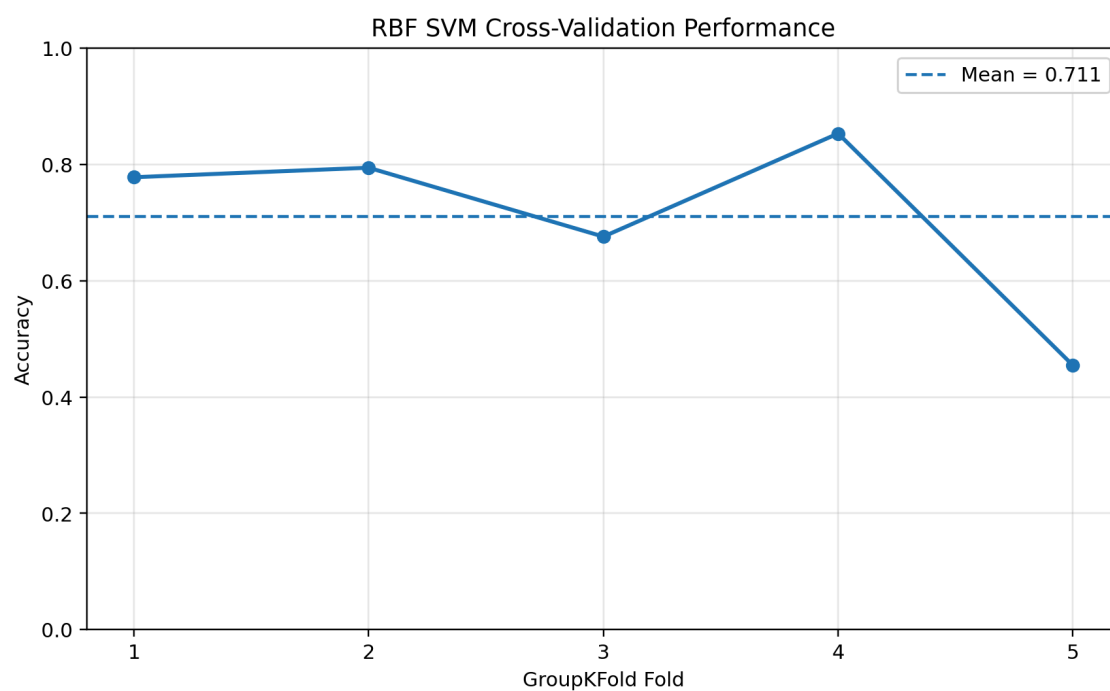
Final PCA + RBF SVM — Window-Level Test Confusion Matrix



B.14 Final window-level PCA-SVM confusion matrix.



B.15 Final recording-level PCA-SVM confusion matrix.



B.16 Fold-wise cross-validation stability.

Appendix C. Project Reproducibility and Supplementary Materials

Four principal files accompany the report: the original CSI_Activity_Recognition.ipynb notebook, csi_receiver.zip containing the CSI dataset and supporting project files, the official IS3IP-2026 report template, and the preliminary WiFi-DensePose technical study.

The complete reproducibility package consists of the original CSI_Activity_Recognition.ipynb notebook, the csi_receiver project archive containing the fifteen activity recordings and supporting source files, the official IS3IP-2026 report template and the preliminary WiFi-DensePose technical study.

C.1 Core Algorithms

The core reproducibility pipeline consists of exact 130-value CSI packet validation; I/Q extraction into a 62×2 representation; amplitude calculation using $\sqrt{I^2 + Q^2}$; 100-packet windows with a 50-packet step; training-only subcarrier normalization; seven statistical features for each of 62 subcarriers; recording-aware GroupKFold model comparison; and the selected StandardScaler $\rightarrow$ PCA(10) $\rightarrow$ RBF-SVM pipeline with $C = 100$ and $\text{gamma} = 0.001$. Final evaluation is reported at both the window and recording levels.

Where the uploaded notebook contained no stored output for a cell, the result was regenerated from the supplied dataset rather than inferred.

